

Senior High School



Lesson Exemplar in General Mathematics

Quarter 3

Unit

9

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Lesson Exemplar for General Mathematics
Quarter 3: Unit 9

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Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR				
Learning Area:	GENERAL MATHEMATICS			Grade Level: 11
Semester:	SECOND	Quarter:	THIRD	Unit: 9

I. LEARNING GOALS	
Content Standard	The learners demonstrate knowledge and understanding of random variables and sampling. Contextualized: <i>The learners demonstrate knowledge and understanding of random variables and sampling in everyday situations and activities, enabling them to make informed decisions and interpret data effectively.</i>
Performance Standard	Learners propose statistical investigation identifying and justifying the sample size and sampling techniques; and draw conclusions from given data using normal distribution and expected values. Contextualized: <i>Learners are able to design and conduct simple statistical studies such as surveys on learner habits or preferences, by selecting suitable samples, calculating expected values based on appropriate sample size, applying normal distribution, and presenting their findings clearly for decision-making.</i>
Learning Competencies	The learners... 1. differentiate between discrete and continuous random variables and describe their uses. 2. calculate the expected value, variance, and standard deviation of discrete random variables. 3. illustrate a normal random variable and its characteristics. 4. convert a normal random variable to a standard normal variable and vice versa. 5. compute probabilities and percentiles involving standard normal distribution using a Z-distribution table or appropriate technologies. 6. solve problems involving normal distribution. 7. illustrate random sampling. 8. determine appropriate sample size for a specific investigation.

	9. distinguish between: - parameter and statistic. - probability and nonprobability sampling techniques.
II. REFERENCES and MATERIALS	Textbook and Modules Lane, D. M. (2017). Normal distribution [PDF]. Rice University. Online Statistics Education: A Multimedia Course of Study. OpenStax. (2019). Introductory statistics. OpenStax CNX. University of West Georgia. (n.d.). The normal distribution notes [PDF]. University of West Georgia. Wasserman, L. (2004). All of statistics: A concise course in statistical inference. Springer. Department of Education. (n.d.). Statistics and probability: Normal distribution (Grade 11, Quarter 3 module). DepEd Philippines. Dela Cruz, A. (2021). Chapter II – Normal distribution [Self-learning module]. Gordon College. (2021). EDUMATH05: Module 6 – Normal distribution [Learning module]. Rene R. Belecina, E. S. (2016). RBS Statistics and Probability (First Edition). In E.S. Rene R. Belecina, RBS Statistics and Probability (First Edition) (pp. 200-210). 856 Nicanor Reyes Sr.ST., Sampaloc, Manila: Rex Book Store, Inc (RBSI). Senior High School Statistics and Probability Modules (DepEd) Websites https://onlinestatbook.com/lms/normal_distribution/normal_distribution.pdf https://assets.openstax.org/oscms-prodcms/media/documents/IntroductoryStatistics-OP_i6tAI7e.pdf https://www.westga.edu/academics/research/vrc/assets/docs/the_normal_distribution_notes.pdf https://www.stat.cmu.edu/~brian/valerie/617-2022/0%20-%20books/2004%20-%20wasserman%20-%20all%20of%20statistics.pdf https://www.studocu.com/ph/document/sti-west-negros-university/general-education/statistics-and-probability-normal-distribution/46299092 https://www.scribd.com/document/555776554/Chapter-II-Normal-Distribution https://www.scribd.com/document/559263316/EDUMATH05-for-online-module-6 https://www.google.com/search?sa=X&sca_esv=e023beae1a6c5125&sxsrf=AE3TifMpEL064jpSIImRffEbVlycA_u_m3A:1757576170674&udm=2&fbs=AlljpHxU7SXXniUZfeShr2fp4giZ1Y6MJ25_tmWITc7uy4KleiAkWG4OIBE2zyCTMjPbGmMU8EWskMk2JSE__efdUJ3xRFvZ0M2vJLB0hUMk5HOE2JYHbn8EkAxflxA6nxAnoThAuIcN7xDcumLPe-3cEEp0BCRKPIuNemU71V5H4BQp-rSd0ykhtsWXyWSGI_xAj-MEbBR2jUQBSnSDsPt34j9ah4jECw&q=normal+curve&ved=2ahUKEwjPjca6mdCPAxW8sFYBHVyeB6MQtKgLegQIFBAB&biw=605&bih=608&dpr=1.5#vhid=WMDtQknxpPXnaM&vssid=mosaic Statistics Lesson 6.3 Notebook. (2017, May 03). Retrived from https://www.hanover.k12.in.us/cms/lib/INO1001361/Centricity/Domain/104/Stats%206.3.pdf

	Video Lessons DepEd TV: Statistics and Probability – Sampling Techniques (https://www.youtube.com/watch?v=5PsF5MsrCOo) Materials • Calculators • Flashcards • z-table • dice • normal distribution curves • colored pens or crayons, pencils • bond papers • graphing papers • Printed activity sheets / scenario cards • Manila paper or chart paper • markers • Slips of paper or name cards (for simulation activities)	
III. CONTENT	Lesson 9.1: Discrete Random Variables (Suggested Time Allotment: 4 hours)	
IV. OBJECTIVES	At the end of the lesson, the learners are expected to: 1. define concepts of random variables, discrete random variables, continuous random variables, expected value, variance, and standard deviation of discrete random variable. 2. differentiate discrete random variables from continuous random variables based on their characteristics 3. identify examples of real-life situations that can be modelled using discrete and continuous random variables. 4. apply formula to compute the expected value, variance, and standard deviation of a discrete random variable.	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge	
	Activity A.1: Leveling Learner Readiness Instructions: 1. Work in pairs. 2. Identify the following variables below as countable or measurable.	Activity A.1: Leveling Learner Readiness In this activity, learners work in pairs to classify different real-life variables as countable or measurable. By observing and discussing each variable, they develop an understanding of how quantities can be quantified—either by simple counting or by using measurement tools. This hands-on approach encourages critical thinking and

Countable means the quantity can be determined by simply counting, without the use of any measuring devices.

Measurable means the quantity must be determined using measuring tools or instruments.

3. Write your answer on the space provided.

Variable	Countable or Measurable
1. Height of a learner	
2. Number of text messages sent	
3. The weight of New born child	
4. Time it takes to finish an examination	
5. Distance from home to school	
6. Number of books in a library	
7. The speed of the moving car	
8. Number of pencils in a box	
9. Number of teachers in a school	
10. Temperature of a room	

Reflection Questions:

1. What features of each variable helped you determine its correct classification?
2. Which group of variables is discrete and which is continuous?
3. Can the values of each variable increase only in whole numbers, or can they include fractions and decimals? What does this tell you about the type of variable?
4. Is it possible for the variable to have infinitely many values within a given range? Why or why not?

prepares learners to recognize patterns in data. It also lays the foundation for the lesson on random variables. By distinguishing between discrete and continuous variables in everyday situations, learners can more easily understand how random variables represent outcomes that depend on chance, which is essential for probability and statistics.

To facilitate this activity, you may apply any of the suggested strategies.

Strategy 1: **Think-Pair-Check**

1. Review the meanings of countable and measurable with simple examples.
2. Ask learners to classify each variable in pairs.
3. Circulate and prompt pairs with questions like: “Can this be counted one by one?” or “Do you need a tool to measure this?”
4. Conduct a whole-class sharing of answers and reasoning.

Strategy 2: **I Do – We Do – You Do**

1. I Do: Model classification for two sample variables on the board.
2. We Do: Classify 2–3 items together with the class, asking guiding questions.
3. You Do: Learners work in pairs to complete the remaining items, then discuss their answers as a class.

After the accomplishing the task, present the reflection questions.

Conclude the activity by saying:

“Now that we have classified variables as countable or measurable, notice how some variables can take only specific values while others can vary continuously. In statistics, we call such variables random variables, which represent outcomes that depend on chance. Today, we will explore how these discrete and continuous random

	5. How would you differentiate a discrete variable from a continuous variable?	variables help us understand and analyze real-life situations.” Answer Key: <table><tr><td>1. Measurable</td><td>6. Countable</td></tr><tr><td>2. Countable</td><td>7. Measurable</td></tr><tr><td>3. Measurable</td><td>8. Countable</td></tr><tr><td>4. Measurable</td><td>9. Countable</td></tr><tr><td>5. Measurable</td><td>10. Measurable</td></tr></table>	1. Measurable	6. Countable	2. Countable	7. Measurable	3. Measurable	8. Countable	4. Measurable	9. Countable	5. Measurable	10. Measurable
1. Measurable	6. Countable											
2. Countable	7. Measurable											
3. Measurable	8. Countable											
4. Measurable	9. Countable											
5. Measurable	10. Measurable											
A.2. Establishing the Purpose of the Lesson												
	Activity A. 2: Appreciating Lesson Relevance Instructions: 1. Form a group of 4–5 members. 2. Your group will be given one real-life situation involving an uncertain outcome. 3. Carefully read the scenario and discuss the questions that follow. 4. Write your group’s answers on a manila paper or in your notebook. 5. Be ready to share your responses with the class. Scenario 1: Grab & Win A box contains folded papers labeled with different prizes such as ₱10, ₱20, ₱50, or “Try Again.” One member will randomly pick a paper. Tasks: 1. List all possible outcomes that can be obtained. 2. Identify which outcomes are more likely and explain why. 3. Represent the prize amount using a variable X. What values can X take?	Activity A. 2: Appreciating Lesson Relevance Learners are placed in small groups and presented with everyday situations where outcomes are uncertain. They must predict possible results and discuss what values could appear. This activity highlights how random variables help us make sense of uncertainty in real life, paving the way for the formal lesson. To facilitate this task, apply one of the suggested strategies. Strategy 1.: Predict–Observe–Explain Instructions: 1. Ask learners to predict what they think will happen before the activity begins. 2. Let them carry out the scenario and observe the results. 3. After the activity, ask the reflection questions. 4. Summarize by emphasizing that such unpredictable outcomes can be represented using numbers called random variables. 5. Introduce the lesson focus using this transition line: “These unpredictable values are what we call random variables. They help us describe and analyze outcomes that depend on chance.”										

	Scenario 2: How Long Will It Take? Two group members will run from the gate to the classroom while another member records the time using a phone timer. Tasks: - Record the time for each run. - Compare the recorded times. Are they exactly the same? Why or why not? - Describe the range of possible values the time can have. Reflection Questions: - Why were you unable to predict the exact outcome even though you knew the situation and possible results? - How do these outcomes show that the values depend on chance or randomness? - How can representing the outcome using a variable help you describe and analyze similar real-life situations?	Strategy 2. Hands-on Exploration Instructions: - Divide the class into groups of 4–5 learners. - Assign each group one scenario (Grab & Win or How Long Will It Take?). - Explain that the task is not to get the “correct” answer but to observe what happens when outcomes depend on chance. - Give groups 5–7 minutes to perform the activity and discuss their observations. - Move around the room and ask probing questions such as: “Did you get the same result every time?” “What made the outcome unpredictable?” - Call on selected groups to share their results and reasoning. - Facilitate the reflection questions. After facilitating this activity, present the learning goals and the performance task to be performed at the end of the unit.									
B. Instituting New Knowledge	B.1. Presenting Examples Activity B.1 Exploring Key Concepts Option 1: Variable Detectives: Classifying the Clues Instructions: Examine and study the table below which shows some variables which are grouped according to their types and reasons. <table> <tr> <th>Variables</th><th>Type</th><th>Reason</th></tr> <tr> <td>Number of children in the park</td><td>Discrete</td><td>Countable, Whole numbers only, no fractions</td></tr> <tr> <td>Weight of luggage</td><td>Continuous</td><td>Measurable with infinite precision</td></tr> </table>		Variables	Type	Reason	Number of children in the park	Discrete	Countable, Whole numbers only, no fractions	Weight of luggage	Continuous	Measurable with infinite precision
Variables	Type	Reason									
Number of children in the park	Discrete	Countable, Whole numbers only, no fractions									
Weight of luggage	Continuous	Measurable with infinite precision									
	Activity B.1 Exploring Key Concepts Option 1: Variable Detectives: Classifying the Clues This activity engages learners in identifying, analyzing, and classifying variables as discrete or continuous through guided examples and collaborative group work. Teachers use this activity to help learners deepen their understanding of how variables behave in real-life contexts by first examining a model table and then applying the concepts to new situations. Learners act as “Variable Detectives,” examining clues (variables), determining their types, explaining their reasoning, and constructing their own examples. The activity encourages critical thinking,										

	Speed of a fighter jet	Continuous	Any value possible within interval, measured	teamwork, and clear communication as learners present and justify their classifications. It also prepares them for later lessons involving probability distributions, sampling, and statistical investigations. Instructions: 1. Introduce the purpose of the activity to help learners distinguish discrete and continuous variables. 2. Present the sample table of variables for learners to observe. 3. Ask learners guiding questions about what is being counted or measured and whether decimals are possible. 4. Instruct learners to form groups of five and discuss their observations. 5. Distribute the printed worksheets with new variables for classification. 6. Direct each group to classify every variable as discrete or continuous. 7. Require groups to write clear reasons for each classification and reflect on their thought process. 8. Ask each group to present their classifications to the class and explain their reasoning. 9. Facilitate discussion by asking probing questions to encourage reflection on differences between discrete and continuous variables. 10. Instruct groups to create ten discrete and ten continuous variable examples with reasons and reflect on how these relate to real-world data collection and research.
	Number of dogs in the neighborhood	Discrete	Countable with no intermediate values	
	Temperature in Baguio City	Continuous	Smooth range of values, infinitely many possibilities	
	Time to finish a 100-meter dash	Continuous	Any fractional value possible, infinitely many values	
	Height of a basketball player	Continuous	Infinite possible values within a range, measurable with decimals	
	Distance from church to school	Continuous	Infinite possible values within a range, measurable with decimals	
	Number of Senior citizens in a barangay	Discrete	Countable whole numbers and finite	
	Number of teachers in school	Discrete	Countable, Whole numbers only, no fractions	
2. Answer the following questions: • In your own words, what does 'discrete variable' mean? • Can you explain what a continuous variable is? • How do discrete and continuous variables differ? 3. Continue by asking the learners to form 5 members in each group. 4. Printed worksheets with variables will be given by your teacher in each group. 5. Classify each variable as discrete or continuous and write your reasons. 6. Your group will present your classification with your reasons as to how you came up with your answers with the class. 7. Come up with your own ten (10) examples of discrete and ten (10) continuous variables. Identify the reasons why it is discrete or continuous.				

Reflection Question:

1. In what ways are you distinguishing discrete variables from continuous variables?
2. In what ways are you connecting the importance of understanding variables to real-world data collection and research?

Option 2:**Exploring Discrete and Continuous Random Variable****Instructions:****1: Study the Table**

Analyze the information in the tables below. Notice the **situation**, the **random variable (X)**, and the **possible values of X**.

Discrete Random Variables

Situation / Experiment	Random Variable (X)	Possible Values of X
Rolling a fair six-sided die	Number that appears on the die	{1, 2, 3, 4, 5, 6}
Tossing 3 coins	Number of heads	{0, 1, 2, 3}
Counting learners in a class	Number of learners present	{0, 1, 2, 3, ..., 50}
Customer arrivals in a shop	Number of customers entering per hour	{0, 1, 2, 3, ...}
Exam results	Number of correct answers out of 20	{0, 1, 2, ..., 20}

Continuous Random Variables

Situation / Experiment	Random Variable (X)	Possible Values of X
Measuring height	Height of a learner (cm)	Any value in a range (e.g., 140.5–190.2)

Option 2:**Exploring Discrete and Continuous Random Variable**

In this activity, learners examine real-life situations and identify the corresponding random variables. They classify each variable as discrete (countable) or continuous (measurable) based on its possible values. By analyzing patterns and reasoning through examples, learners develop a clear understanding of the differences between discrete and continuous random variables. This hands-on exploration helps learners connect abstract statistical concepts to everyday experiences and prepares them for further study of random variables in probability and statistics.

Instructions:

1. Introduce the activity by explaining that a random variable is a numerical quantity whose value depends on chance and stating the goal of classifying variables as discrete or continuous.
2. Guide learners' observation by asking them to examine each situation and prompting them with questions such as what is being counted or measured, whether the values can be listed one by one, and whether they include decimals or only whole numbers.
3. Let learners work in pairs or small groups to identify the random variable in each scenario and classify it as discrete or continuous while you circulate and ask probing questions to check reasoning.
4. Facilitate a brief class discussion where groups share their classifications and highlight key ideas such as

Recording travel time	Time to go to school (minutes)	Any positive real number
Measuring weight	Weight of a person (kg)	Any value in a range
Measuring temperature	Temperature outside (°C)	Any value in a range
Measuring voltage	Electrical voltage (V)	Any value in a range

Classify the Variables

1. Work in pairs or small groups.
2. For each situation in the tables, identify:
 - o The **random variable**
 - o Whether it is **discrete** or **continuous**
3. Write down your classification and be ready to explain your reasoning.

Accomplish Reflection Questions

Answer individually or discuss with your group:

1. What clues helped you identify whether a variable is discrete or continuous?
2. Can you give a real-life example of each type of random variable (not in the table)?
3. How would you explain the difference between discrete and continuous random variables to a friend?

discrete variables being countable whole numbers and continuous variables being measurable values that can take any number in a range.

5. End with a short reflection where learners answer questions individually or in groups, followed by a summary of key points and a connection to the upcoming lesson on random variables in probability and statistics.

B.2. Discussing the Concept

Activity B.2: Deepening Conceptual Understanding

Instructions:

1. Carefully read the lesson notes on sampling.
2. Take note of important terms.
3. Work through the step-by-step examples provided
4. Be ready to answer questions or share your thoughts during discussions.
5. Raise questions if their concepts are confusing.

Activity B.2: Deepening Conceptual Understanding

In this part of the lesson, learners explore the concept of random variables and apply it to real-life situations and experiments. They will distinguish between discrete and continuous random variables, identify the possible outcomes and values, and calculate key measures for discrete random variables, including expected value, variance, and standard deviation. Through discussion, examples, and guided problem-solving, learners develop the skills to analyze uncertainty and variability in real-world events, from games to surveys to everyday measurements.

Random Variables

A **random variable** is a variable whose value is determined by chance. It may take on numerical or categorical (nominal) values, depending on the experiment. Random variables are usually denoted by capital letters such as X, Y, or Z, and they represent the set of possible outcomes of a random experiment.

Example: 1.

A coin is tossed 3 times. Let X be the random variable denoting the number of heads. Find the values of the random variable X.

Solution:

To accomplish this, first identify all possible outcomes of the given statistical experiment, and then determine the specific random variable defined in the problem, i.e.

Outcomes	TTT	TTH	THT	THH	HTT	HTH	HHT	HHH
x=no. of heads	0	1	1	2	1	2	2	3

Therefore, the possible values of the random variable X are 0, 1, 2 and 3.

Example 2:

The random variable Y denotes the **number of even number outcomes** in a roll of a die. Find the possible values of Y.

Die Outcome	Is it Even?	Y = Number of Even Numbers
1	No	0
2	Yes	1
3	No	0
4	Yes	1

Prior to the discussion be reminded on the following:

1. Given the number of concepts to be covered, carefully plan your instructional approach to ensure learners can engage with the lesson without feeling overwhelmed.
2. Use visual aids like graphs and shaded areas to strengthen understanding.
3. Encourage learners to verbalize reasoning and explain steps to peers.
4. Integrate technology where possible.
5. Continuously relate abstract concepts to tangible, familiar examples.
6. The strategies provided are merely suggestions; you may apply the approach you believe best suits your learners during the discussion.
7. Ask check-in questions every concept and reflection question every end of the lesson.

To facilitate the discussion, apply any of the suggested strategies.

Strategy 1: Think-Pair-Share

Instructions:

1. Present a random variable example.
2. Ask learners to think individually about the possible outcomes and values.
3. Pair learners to compare answers and discuss reasoning.
4. Share and discuss answers as a class, emphasizing the identification of discrete vs. continuous variables and the calculation of expected value, variance, and standard deviation.

Strategy 2: Guided Problem Solving

Instructions:

1. Introduce a problem.

5	No	0
6	Yes	1

From the table, Y can take the values (0, 1).

More examples of random variables:

1. The number of getting hearts in drawing five cards from a standard deck, the number of hearts drawn (0, 1, 2, 3, 4, or 5).
2. Number getting heads or tails when flipping a coin. (0,1)
3. The number obtained when rolling a die (1, 2, 3, 4, 5, or 6).
4. The number of head that will appear in tossing a coin three times. (0, 1, 2, 3)
5. The sum of dots in rolling a pair of dice.

The importance of random variables lies in their ability to help us understand and predict uncertain real-world events. Each random variable has an underlying probability distribution, which specifies the likelihood of each possible value occurring. This allows us to estimate probabilities of different outcomes and make informed decisions based on this information.

Random variables are widely used in fields where risk assessment and decision-making are crucial, such as finance, engineering, and medicine. For example, in finance, random variables model investment returns, helping investors allocate resources wisely and manage risk effectively.

Types of Random Variables

A **discrete** random variable is a type of variable that takes values you can count.

2. Work through step-by-step calculations with the class, prompting learners to:
3. Identify the random variable X and its possible values.
4. Compute the expected value ($E(X)$).
5. Calculate variance and standard deviation.
6. Discuss the interpretation of the results in context.
7. Encourage learners to solve similar examples individually or in small groups, then discuss results as a class.

Additional Tips

1. There are many concepts to be discussed, manage the lesson carefully.
2. Use the prompt questions during discussions.
 - “What is a random variable, and how do we know if it is discrete or continuous?”
 - “How do we determine the possible values of a random variable?”
 - “Why is expected value important, and what does it tell us about the outcomes?”
 - “How do variance and standard deviation help us understand risk or spread in real-life situation
3. For check-ins, ask the following questions:
 - How did identifying the random variable help you understand what outcomes were being measured or counted in each situation?
 - Why is it important to know the expected value, variance, and standard deviation when making decisions in real-life situations involving chance or risk?
 - How can the concept of random variables help you better interpret data from games, surveys, or everyday experiences?

Example 3.

1. The number of coins in your pocket.
2. The number of learners in a class.
3. The number of calls received at a help desk
4. The number of goals scored in a basketball game.
5. The number of customers who enter a store.

A **continuous** random variable is one whose values are obtained through measurement.

Example 4:

1. The time it takes for a runner to finish a race.
2. The height of each learner in a classroom.
3. The exact amount of milk in a carton.
4. The temperature recorded at noon in a city.
5. The volume of water in a swimming pool.

Probability of a Discrete Random Variable

A **discrete random variable** is one that can take on countable values (e.g., 0, 1, 2, ...). Each possible value has an associated probability, which satisfies:

1. $0 \leq P(X = x_i) \leq 1$ for each value x_i
2. $\sum_i P(X = x_i) = 1$

The probability distribution of a discrete random variable lists all possible values and their probabilities.

Graph of a Probability Distribution of a Discrete Random Variable

A **graph of a probability distribution** for a discrete random variable is a **probability histogram** or a **stick graph** (vertical line graph) that shows each possible value of the variable along with its probability.

Because the variable is *discrete*, the graph has **separate bars or lines**, not a continuous curve.

How to Graph a Discrete Probability Distribution

1. List the values of the random variable (e.g., 0, 1, 2, 3 heads).
2. List the probability of each value.
3. Draw the x-axis labeled with the values of the random variable.
4. Draw the y-axis labeled with "Probability".
5. Draw bars or vertical lines whose heights correspond to the probability of each value.
6. Make sure the bars do not touch, since the data is discrete.

Example 5.

Construct a probability distribution showing the number of heads that will appear when tossing a coin three times and draw the graph.

Solutions:

Step 1: Identify the random variable

Let X = number of heads in 3-coin tosses.

Possible values of X are:

0,1,2,3

Step 2: List all possible outcomes

There are $2^3 = 8$ outcomes when tossing a coin three times:

TTT, TTH, THT, THH, HTT, HTH, HHT, HHH

Step 3: Count the number of heads for each outcome

Outcome	Number of Heads (X)
TTT	0
TTH	1
THT	1
THH	2

HTT	1
HTH	2
HHT	2
HHH	3

Step 4: Determine probabilities

X (Number of Heads)	Number of Outcomes	Probability P(X)
0	1	$1/8 = 0.125$
1	3	$3/8 = 0.375$
2	3	$3/8 = 0.375$
3	1	$1/8 = 0.125$

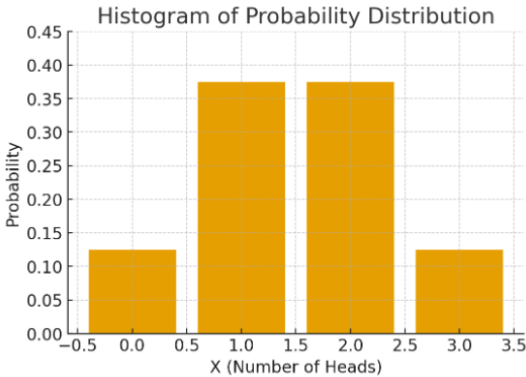
Step 5: Probability Distribution Table

Heads (X)	0	1	2	3
P(X)	0.125	0.375	0.375	0.125

Check:

$0.125 + 0.375 + 0.375 + 0.125 = 1✓$

Graph:



Example 6.

Construct a probability distribution table showing the probability of the number of even numbers in a single die roll draw the graph.

Solutions:

Step 1: Define the random variable

Let Y = number of even numbers in one roll of a standard six-sided die.

Possible values of Y are: **0, 1**

Step 2: List possible outcomes

Die Outcome	Is it Even?	Y
1	No	0
2	Yes	1
3	No	0
4	Yes	1
5	No	0
6	Yes	1

Step 3: Determine probabilities

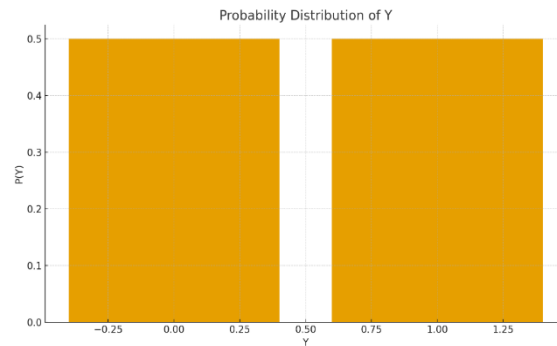
Y (Number of Even Numbers)	Number of Outcomes	Probability P(Y)
0	3	$3/6 = 0.5$
1	3	$3/6 = 0.5$

Step 4: Probability Distribution Table

Y	P(Y)
0	0.5
1	0.5

Check:

$$0.5 + 0.5 = 1\checkmark$$

Graph:**Expected Value**

The **expected value** or **mean** (denoted as $E(X)$ or μ) of a random variable X is the average or mean value that $\mathbf{X}$ will take over many repetitions of an experiment. It represents the long-run average outcome.

Expected value of Discrete Random Variable

For a discrete random variable with values $x_1, x_2, x_3, \dots, x_n$ and probabilities $p_1, p_2, p_3, \dots, p_n$:

$$E(X) = \sum X \cdot P(X)$$

where:

$E(X)$ = expected value or mean

X = random variable

$P(X)$ = probability of the random variable.

Example 7.

A fair six-sided die is rolled. Let X be the number that appears. Find the expected value of X .

Solutions:

A fair die has outcomes: 1, 2, 3, 4, 5,

$$E(X) = \sum X \cdot P(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6}$$

$$E(X) = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5$$

Answer: $E(X) = 3.5$

Example 8.

A game costs ₦10 to play. A player draws a card from a box with 5 cards: 2 cards give ₦0, 2 cards give ₦20, and 1 card gives ₦50. Let X be the net gain (prize minus cost). Find the expected value of the game.

Solutions:

Possible net gains:

₦0 prize $\rightarrow X = -10$ (probability $2/5$)

₦20 prize $\rightarrow X = 10$ (probability $2/5$)

₦50 prize $\rightarrow X = 40$ (probability $1/5$)

$$E(X) = (-10) \cdot \frac{2}{5} + 10 \cdot \frac{2}{5} + 40 \cdot \frac{1}{5} = -4 + 4 + 8 = 8$$

Answer: Expected gain = ₦8.00

Example 9.

A survey shows the number of books learners read in a month as follows. Find the expected number of books a learner reads.

Number of books (X)	0	1	2	3	4
Probability P(X)	0.1	0.2	0.4	0.2	0.1

Solutions:

$$E(X) = \sum X \cdot P(X) = 0 \cdot 0.1 + 1 \cdot 0.2 + 2 \cdot 0.4 + 3 \cdot 0.2 + 4 \cdot 0.1$$

$$E(X) = 0 + 0.2 + 0.8 + 0.6 + 0.4 = 2$$

Answer: Expected number of books = 2

Variance and Standard Deviation of Discrete Random Variable

Variance is a measure of how much the values of a random variable deviate from the mean (expected value). It gives an idea of the spread or dispersion of the data. For a discrete random variable X with expected value $E(X)$, the variance is calculated as:

$$\mathbf{Var}(X) = \sum (X - E(X))^2 \cdot P(X)$$

X = possible value of the random variable

$E(X)$ = expected value (mean) of X

$P(X)$ = probability of X

A higher variance indicates that the values are more spread out from the mean, while a lower variance indicates that they are closer to the mean.

Standard deviation is the square root of the variance. It is expressed in the same units as the random variable, making it easier to interpret.

$$\mathbf{SD}(X) = \sqrt{\mathbf{Var}(X)}$$

Like variance, a higher standard deviation means the data is more spread out, while a lower standard deviation means the data is more concentrated around the mean.

In short:

Variance measures spread in squared units.

Standard deviation measures spread in original units, making it easier to understand and compare.

Example 10.

Determine the variance and standard deviation of example 7, 8, and 9.

Solutions:*Variance and Standard Deviation of Example 7***Problem Recap:**

X = number appearing on a fair six-sided die.

Step 1. **Expected Value (Mean)**

$$E(X) = 3.5$$

Step 2. **Variance Formula**

$$\text{Var}(X) = \sum (X - E(X))^2 \cdot P(X)$$

Step 3. **Calculate Variance**

X	P(X)	X - E(X)	(X - E(X)) ²	(X - E(X)) ² · P(X)
1	1/6	-2.5	6.25	6.25 × 1/6 ≈ 1.0417
2	1/6	-1.5	2.25	2.25 × 1/6 ≈ 0.375
3	1/6	-0.5	0.25	0.25 × 1/6 ≈ 0.0417
4	1/6	0.5	0.25	0.25 × 1/6 ≈ 0.0417
5	1/6	1.5	2.25	2.25 × 1/6 ≈ 0.375
6	1/6	2.5	6.25	6.25 × 1/6 ≈ 1.0417

$$\begin{aligned}\text{Var}(X) &= 1.0417 + 0.375 + 0.0417 + 0.0417 + 0.375 + 1.0417 \\ &= 2.9168 \approx 2.92\end{aligned}$$

Step 4. **Standard Deviation**

$$SD(X) = \sqrt{\text{Var}(X)} = \sqrt{2.9168} \approx 1.71$$

Answer: Variance ≈ 2.92, SD ≈ 1.71*Variance and Standard Deviation of Example 9***Problem Recap:**

Number of books (X)	0	1	2	3	4
Probability P(X)	0.1	0.2	0.4	0.2	0.1

Net gain X = -10, 10, 40 with probabilities 2/5, 2/5, 1/5

Step 1. **Expected Value**

$$E(X) = (-10) \cdot \frac{2}{5} + 10 \cdot \frac{2}{5} + 40 \cdot \frac{1}{5} = 8$$

Step 2. **Variance Formula**

$$\text{Var}(X) = \sum (X - E(X))^2 \cdot P(X)$$

Step 3. **Calculate Variance**

X	P(X)	X - E(X)	(X - E(X)) ²	(X - E(X)) ² · P(X)
-10	2/5	-18	324	324 × 2/5 = 129.6
10	2/5	2	4	4 × 2/5 = 1.6
40	1/5	32	1024	1024 × 1/5 = 204.8

$$\text{Var}(X) = 129.6 + 1.6 + 204.8 = 336$$

Step 4. **Standard Deviation**

$$SD(X) = \sqrt{336} \approx 18.33$$

Answer: Variance = 336, SD ≈ 18.33

Variance and Standard Deviation of Example 9

Step 1. **Expected Value**

$$E(X) = 0 * 0.1 + 1 * 0.2 + 2 * 0.4 + 3 * 0.2 + 4 * 0.1 = 2$$

Step 2. **Variance Formula**

$$\text{Var}(X) = \sum (X - E(X))^2 \cdot P(X)$$

Step 3. **Calculate Variance**

X	P(X)	X - E(X)	(X - E(X)) ²	(X - E(X)) ² · P(X)
0	0.1	-2	4	4 × 0.1 = 0.4
1	0.2	-1	1	1 × 0.2 = 0.2

2	0.4	0	0	$0 \times 0.4 = 0$
3	0.2	1	1	$1 \times 0.2 = 0.2$
4	0.1	2	4	$4 \times 0.1 = 0.4$

$$\text{Var}(X) = 0.4 + 0.2 + 0 + 0.2 + 0.4 = 1.2$$

Step 4: Standard Deviation

$$SD(X) = \sqrt{1.2} \approx 1.095 \approx 1.10$$

Answer: Variance = 1.2, SD $\approx$ 1.10

Data Analysis Using Expected Value, Variance and Standard Deviation

Example 11.

A player plays a simple board game where they can win or lose money based on the roll of a spinner. The outcomes are:

X (Net Gain in ₱)	-20	0	10	30
P(X)	0.2	0.3	0.4	0.1

Find the expected value, variance, and standard deviation of the net gain. Interpret the results.

Solutions:

Step 1. Expected Value (Mean)

$$\begin{aligned}
 E(X) &= \sum X \cdot P(X) \\
 &= (-20)(0.2) + (0)(0.3) + (10)(0.4) \\
 &\quad + (30)(0.1) \\
 E(X) &= -4 + 0 + 4 + 3 = 3
 \end{aligned}$$

Interpretation:

On average, a player can expect to win ₱3 per game.

Step 2. Variance

$$\text{Var}(X) = \sum (X - E(X))^2 \cdot P(X)$$

X	P(X)	X - E(X)	(X - E(X)) ²	(X - E(X)) ² · P(X)
-20	0.2	-23	529	$529 \times 0.2 = 105.8$

0	0.3	-3	9	$9 \times 0.3 = 2.7$
10	0.4	7	49	$49 \times 0.4 = 19.6$
30	0.1	27	729	$729 \times 0.1 = 72.9$

$$\text{Var}(X) = 105.8 + 2.7 + 19.6 + 72.9 = 201$$

Step 3. **Standard Deviation**

$$SD(X) = \sqrt{\text{Var}(X)} = \sqrt{201} \approx 14.18$$

Step 4. **Interpretation**

Expected value: On average, the player wins ₱3.)) per game.

Variance: The variance of 201 tells that there is a high spread in possible outcomes, meaning results can vary widely.

Standard deviation: $\approx$ The standard deviation of ₱14.18 implies that most of the game outcomes typically deviate around ₱14.00 from the mean, so the game is quite risky.

It can conclude that, while the player expects a small profit on average, the large standard deviation indicates high variability, meaning they could either lose significantly or win much more than the average.

Remember:

Understanding random variables and how to measure their average and variability helps us make sense of uncertainty in everyday life. By applying these concepts, we can better analyze situations, make informed decisions, and even predict outcomes in games, experiments, and real-world events.

B.3. Developing Mastery

Activity B.3: Practicing Learned Skills

Instructions:

Activity B.3: Practicing Learned Skills

This practice exercise allows learners to apply their understanding of random variables in real-life situations by

Perform the indicated tasks. Write your answer on a sheet paper.

A. Identifying Discrete or Continuous Random Variables

1. Identify whether each random variable is Discrete (D) or Continuous (C).
2. Number of learners absent in a class
3. Time (in minutes) spent answering a test
4. Number of cars passing a toll booth in an hour
5. Weight of a newborn baby
6. Number of defective items in a box

B. Construct probability distribution table for each probability experiment and draw the graph.

1. Number of heads in tossing a coin twice.
2. Getting odd numbers in rolling a die.
3. Number of tails in tossing three coins.
4. Getting a heart when drawing a card from standard deck.
5. Sum of numbers on the two dice.

C. Find the mean, variance, and standard deviation of each problem, then interpret the results.

1. A learner buys snacks during recess. The number of snacks she buys per day (X) follows this distribution:

Number of snacks (X)	0	1	2
P(X)	0.2	0.5	0.3

2. A tricycle driver observes the number of passengers picked up in one trip.

Passengers (X)	1	3	5
P(X)	0.4	0.4	0.2

3. A customer service agent records the number of calls received in one hour.

Calls (X)	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

identifying whether a variable is discrete or continuous and calculating the mean (expected value), variance, and standard deviation for discrete random variables. Learners will interpret their results to understand the typical outcomes and variability in everyday scenarios such as school, games, surveys, and community events. This activity promotes critical thinking, data analysis, and decision-making skills by connecting mathematical concepts to practical applications.

Instructions:

1. Briefly review discrete vs. continuous random variables and the formulas for mean, variance, and standard deviation.
2. Learners solve the exercises individually or in small groups, identifying the random variable, classifying it, and computing the required measures.
3. Discuss each problem step-by-step. Ask prompts such as: "How did you identify the random variable?"; "Why is it discrete?"; "What does the mean tell us?"; "How does the standard deviation describe variability?"
4. Learners share interpretations of results. Emphasize real-life applications and the importance of understanding expected outcomes and variability.

Suggested Strategies:

- **Think-Pair-Share:** Learners solve individually, then compare answers with a partner, and finally share with the class.
- **Guided Problem Solving with Class Discussion:** Work through one problem as a class, guiding learners step-by-step before they attempt similar problems independently.

Answer Key:

Part 1.

1.	D	2.	C	3.	D	4.	C	5.	D
----	---	----	---	----	---	----	---	----	---

A

typically handles about 2 calls per hour, with relatively low fluctuation.

4. The number of attempts a learner needs to pass an online quiz is recorded.

Attempts (X)	2	4	6
P(X)	0.5	0.3	0.2

5. A barangay health worker records the number of family members who received a flu vaccine in a day. The probability distribution is:

Family Members Vaccinated (X)	0	1	2	3
P(X)	0.1	0.4	0.3	0.2

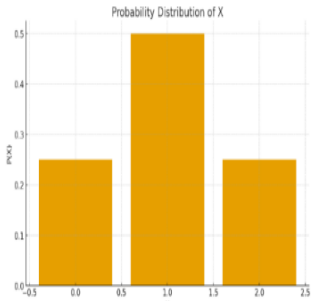
D. Answer the following reflection questions:

1. How did identifying the random variable and its type help you understand the problem better?
2. What does the mean, variance, and standard deviation tell you about the expected outcomes and variability in each scenario?
3. How can understanding random variables and their measures help you make better decisions in real-life situations involving uncertainty?

Part B

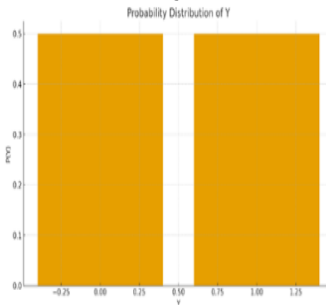
1.

X	P(X)
0	0.25
1	0.50
2	0.25



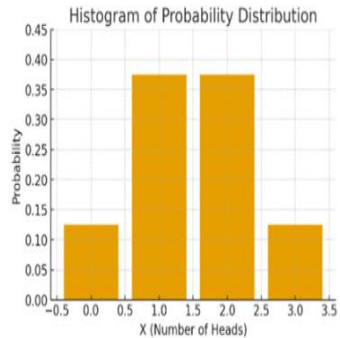
2.

Y	P(Y)
0	0.5
1	0.5



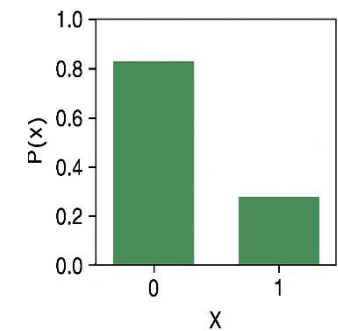
3.

X	P(X)
0	0.125
1	0.375
2	0.375
3	0.125



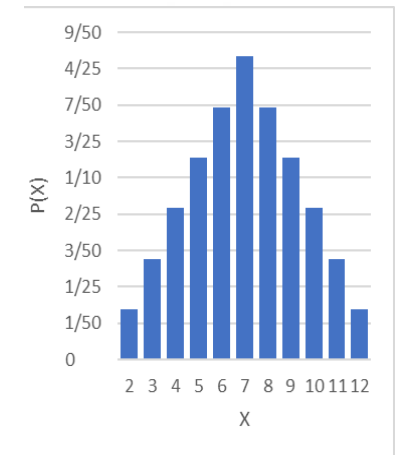
4.

X	P(X)
0	0.75
1	0.25



5.

X	P(X)
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36



Part
C

1. Mean = 1.1 snacks
Variance = 0.49
Standard Deviation = 0.70
Interpretation: On average, the learner buys about 1 snack per day, with small variation in the number purchased.
2. Mean = 2.6 passengers
Variance = 2.44
Standard Deviation = 1.56
Interpretation: The driver expects about 3 passengers per trip, but the number varies noticeably

		3. Mean = 1.7 calls Variance = 0.81 Standard Deviation = 0.90 Interpretation: The agent 4. Mean = 3.4 attempts Variance = 2.64 Standard Deviation = 1.62 Interpretation: Learners generally pass after about 3–4 attempts, but performance varies moderately. 5. Mean = 1.6 family members Variance = 0.84 Standard Deviation = 0.92 Interpretation: On average, about 2 family members									
C. Demonstrating Knowledge and Skills	C.1 Finding Practical Application										
	Activity C.1: Making Real-World Connections	Activity C.1: Making Real-World Connections									
	Instructions: Analyze the scenario and perform the indicated task. Scenario: Your school canteen wants to improve its service by stocking the snacks and drinks that learner like the most. A survey will be conducted to find out which snack and drink combination learners prefer. The combinations to be surveyed are: <table><tr><th>Snack</th><th>Drink</th></tr><tr><td>Chips</td><td>Juice</td></tr><tr><td>Sandwiches</td><td>Water</td></tr><tr><td>Fruits</td><td>Milk</td></tr><tr><td>Cookies</td><td>Soda</td></tr></table>	Snack	Drink	Chips	Juice	Sandwiches	Water	Fruits	Milk	Cookies	Soda
Snack	Drink										
Chips	Juice										
Sandwiches	Water										
Fruits	Milk										
Cookies	Soda										

	The random variable (X) in this investigation is the number of learners who prefer each snack and drink combination. This is a discrete random variable because it can be counted. You will conduct survey to a sample of 20–30 classmates, record their preferred snack and drink combination, and organize the results in a table. Then, they will calculate the expected value (mean), variance, and standard deviation of the number of learners preferring each combination to determine which combination is most popular and how much variation exists among learners’ preferences. Finally, you will interpret the results and make recommendations to the canteen about which snack and drink combinations to stock more or less based on the data collected.	- Guide learners in selecting a sample (20–30 learners) and choosing an appropriate sampling method (simple random, stratified, or cluster). - Learners survey classmates and record results in the table. - Learners compute the expected value (mean), variance, and standard deviation for each snack and drink combination. - Learners interpret the results and make recommendations for the canteen. Encourage them to explain reasoning clearly. - For check-ins, ask the following reflection questions: - How did calculating the mean, variance, and standard deviation help you understand learners’ snack and drink preferences? - Which snack and drink combination was most preferred, and how does the variability affect your recommendation? - How can understanding random variables and their measures help you make better decisions in real-life situations?
C.2. Making Generalization		
	Activity C.2: Wrapping-up the Lesson Option 1: Complete the Rule Instructions: Fill in the missing ideas to complete the generalization statements: - A probability distribution lists all possible values of a random variable and their _____. - The sum of all probabilities in a probability distribution must equal _____. - The expected value represents the long-term or average outcome and is computed by the formula: _____.	Activity C.2: Wrapping-up the Lesson To wrap-up the lesson, you may choose any of the suggested activities. Option 1: Complete the Rule This activity helps learners summarize what they discovered about probability distributions, expected value, variance, and standard deviation by identifying patterns and completing key statements. Instructions: Distribute the “Complete the Rule” card or write the prompts on the board.

	4. The variance measures how spread out the values are from the mean and is computed using: _____. 5. The standard deviation is the _____ of the variance. Option 2: 3–2–1 Quick Wrap-Up Instructions: Write your responses to the following: • 3 – Three things you learned (Example: how to construct a probability distribution, what expected value means, how variance shows variability.) • 2 – Two connections you made (Example: how probability affects expected value; how variance and standard deviation relate.) • 1 – One question you still have (Anything you still find confusing or want to explore more.)	2. Ask learners to work individually first, then confirm answers with a partner. 3. Facilitate sharing of answers and guide learners toward the correct generalizations. 4. Highlight and clarify any misconceptions that appear during the sharing. Option 2: 3–2–1 Quick Wrap-Up This activity guides learners to synthesize their understanding by identifying key ideas, important relationships, and lingering questions about probability distributions and discrete random variables. Instructions: 1. Ask learners to take a half sheet of paper or open their notebook. 2. Read the prompts one by one and give learners time to respond. 3. Invite a few volunteers to share their answers. 4. Highlight patterns in their responses to reinforce the main concepts. 5. Address any common misconceptions that emerge.
C.3. Evaluating Learning		
	Activity C.3: Assessing Learning Outcomes Demonstrating Understanding on Random Variables Instructions: Do the indicated task. Write your answer on a separate sheet of paper. Problem 1. Scenario: A coin is tossed twice. Let X = number of heads. • Construct the probability distribution of X. • Find the expected value, variance, and standard deviation of X.	Activity C.3: Assessing Learning Outcomes Demonstrating Understanding on Random Variables This activity allows learners to demonstrate their understanding of discrete random variables by constructing probability distribution tables and calculating expected value, variance, and standard deviation. It provides an opportunity for learners to apply their knowledge to different scenarios, interpret the results, and reflect on their confidence and perception of statistical concepts.

Problem 2.

Scenario: A fair six-sided die is rolled once. Let $Y = 1$ if an even number occurs, 0 if an odd number occurs.

- Construct the probability distribution of Y .
- Compute the expected value, variance, and standard deviation of Y .

Problem 3.

Scenario: A player draws a card from a standard deck of 52 cards. Let $X = 1$ if the card is a face card (Jack, Queen, King), 0 otherwise.

- Construct the probability distribution of X .
- Calculate the expected value, variance, and standard deviation.

Problem 4.

Scenario: A learner answers a multiple-choice question with 4 options. Let $X = 2$ points if correct, 0 if incorrect. The probability of answering correctly is 0.25.

- Construct the probability distribution of X .
- Find the expected value, variance, and standard deviation of the points earned.

Problem 5.

Scenario: A small lottery game offers the following prizes:

₱1000 with probability 0.05

₱500 with probability 0.10

₱100 with probability 0.20

₱0 with probability 0.65

Let X = winnings per ticket.

- Construct the probability distribution of X .
- Determine the expected value, variance, and standard deviation.
- Interpret the results in terms of average winnings and risk.

The instructions below will serve as your guide in facilitating this activity.

Instructions:

1. Distribute the set of problems to learners.
2. Ask learners to **construct probability distribution tables** for each scenario and calculate expected value, variance, and standard deviation.
3. Observe learners carefully to ensure they understand each step, offering **clarifying questions** rather than answers.
4. Facilitate a brief discussion where learners **share their results and interpretations**.
5. Encourage learners to reflect on their **confidence and perception** regarding their ability to apply these statistical concepts correctly.

Answer Key

1	X	0	1	2
	P(X)	0.25	0.50	0.25

Expected Value, Variance, and Standard Deviation:

$$E(X) = 1, Var(X) = 0.5, SD(X) \approx 0.707$$

2	Y	0	1
	P(Y)	0.5	0.5

$$E(Y) = 0.5, Var(Y) = 0.25, SD(Y) = 0.5$$

3	X	0	1
	P(X)	0.769	0.231

$$E(X) = 0.231, Var(X) = 0.178, SD(X) \approx 0.422$$

4	X	0	2
	P(X)	0.75	0.25

$$E(X) = 0.5, Var(X) = 0.75, SD(X) \approx 0.866$$

5

X	0	100	500	1000
P(X)	0.65	0.20	0.10	0.05

$$E(X) = 95, Var(X) = 82207.5, SD(X) \approx 286.7$$

Interpretation: The expected winnings are ₱95.00, but the high standard deviation indicates large variability, meaning the game is risky.

C.4. Additional Activities

Activity C.4: Extending and Reinforcing Learning

Instructions:

For each scenario:

- Construct the probability distribution table for each scenario.
- Compute Expected Value ($E(X)$), Variance ($Var(X)$), and Standard Deviation ($SD(X)$).
- Interpret your results in context (e.g., risk, variability, or average outcome).

Problem 1.

Scenario: A coin is tossed once. Let $X = 1$ if heads, 0 if tails.

X	0	1
P(X)	0.5	0.5

Problem 2

Scenario: Roll a die. Let $Y = 1$ if number > 4 , 0 otherwise.

Y	0	1
P(Y)	4/6	2/6

Problem 3

Scenario: A team can score 0, 1, or 2 goals in a game with probabilities 0.2, 0.5, 0.3 respectively.

X	0	1	2
P(X)	0.2	0.5	0.3

Activity C.4: Extending and Reinforcing Learning

This activity is designed to give learners additional practice in applying the concepts of expected value, variance, and standard deviation to real-life situations and to deepen their conceptual understanding of discrete and continuous random variables.

Through both individual or group work, learners will strengthen their problem-solving skills, interpret statistical results, and justify their reasoning.

Instructions:

1. Present the additional activities for the learners to answer.
2. Give items 1-5 for learners needing remediation and 6-10 learners that ready for enhancement. This shall be based on the results of evaluation.

Answer Key:

Problem 1

X	0	1
P(X)	0.5	0.5

$$E(X) = 0.5, Var(X) = 0.25, SD(X) = 0.5$$

Problem 2

Y	0	1
P(Y)	$4/6 \approx 0.667$	$2/6 \approx 0.333$

Problem 4

Scenario: A true/false question: $X = 1$ if correct, 0 if incorrect. Probability correct = 0.6.

X	0	1
P(X)	0.4	0.6

Problem 5

Scenario: A shop expects 0, 1, or 2 customers in an hour with probabilities 0.1, 0.6, 0.3.

X	0	1	2
P(X)	0.1	0.6	0.3

Problem 6.

Scenario: Roll two dice. Let X = sum of the dice. Construct probability distribution.

Problem 7.

Scenario: A small store expects daily sales of ₱0, ₱50, ₱100, ₱150 with probabilities 0.1, 0.3, 0.4, 0.2.

Problem 8.

Scenario: A spinner gives net gains: ₱-10 (0.2), ₱0 (0.3), ₱20 (0.4), ₱50 (0.1).

Problem 9

Scenario: A quiz has 3 points max. Probability of scoring 0, 1, 2, 3 points: 0.1, 0.2, 0.4, 0.3.

Problem 10.

Scenario: A factory produces 0, 1, 2, 3 defective items per batch with probabilities 0.05, 0.25, 0.45, 0.25.

$$E(Y) = 0.333, Var(Y) = 0.222, SD(Y) \approx 0.471$$

Problem 3

X	0	1	2
P(X)	0.2	0.5	0.3

$$E(X) = (0 * 0.2) + (1 * 0.5) + (2 * 0.3) = 1.1$$

$$Var(X) = [(0 - 1.1)^2 * 0.2 + (1 - 1.1)^2 * 0.5 + (2 - 1.1)^2 * 0.3] = 0.69$$

$$SD(X) \approx 0.831$$

Problem 4

X	0	1
P(X)	0.4	0.6

$$E(X) = 0.6, Var(X) = 0.24, SD(X) \approx 0.490$$

Problem 5

X	0	1	2
P(X)	0.1	0.6	0.3

$$E(X) = (0 * 0.1) + (1 * 0.6) + (2 * 0.3) = 1.2$$

$$Var(X) = [(0 - 1.2)^2 * 0.1 + (1 - 1.2)^2 * 0.6 + (2 - 1.2)^2 * 0.3] = 0.56$$

$$SD(X) \approx 0.748$$

Problem 6.

X	2	3	4	5	6	7	8	9	10	11	12
P(X)	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

$$E(X) = 7, Var(X) \approx 5.833, SD(X) \approx 2.415$$

Problem 7.

X	0	50	100	150
P(X)	0.1	0.3	0.4	0.2

		$E(X) = 85, Var(X) = 2475, SD(X) \approx 49.75$ Problem 8 <table><tr><td>X</td><td>-10</td><td>0</td><td>20</td><td>50</td></tr><tr><td>P(X)</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.1</td></tr></table> $E(X) = 10, Var(X) = 396, SD(X) \approx 19.90$ Problem 9 <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>0.1</td><td>0.2</td><td>0.4</td><td>0.3</td></tr></table> $E(X) = (0 * 0.1) + (1 * 0.2) + (2 * 0.4) + (3 * 0.3) = 1.9$ $Var(X) = 0.89, SD(X) \approx 0.943$ Problem 10. <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>0.05</td><td>0.25</td><td>0.45</td><td>0.25</td></tr></table> $E(X) = (0 * 0.05) + (1 * 0.25) + (2 * 0.45) + (3 * 0.25) = 1.7$ $Var(X) = 0.905, SD(X) \approx 0.951$	X	-10	0	20	50	P(X)	0.2	0.3	0.4	0.1	X	0	1	2	3	P(X)	0.1	0.2	0.4	0.3	X	0	1	2	3	P(X)	0.05	0.25	0.45	0.25
X	-10	0	20	50																												
P(X)	0.2	0.3	0.4	0.1																												
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X	0	1	2	3																												
P(X)	0.05	0.25	0.45	0.25																												
III. CONTENT																																
Lesson 9.2: The Normal Distribution (Suggested Time Allotment: 6 hours)																																
IV. OBJECTIVES	At the end of the lesson, the learners are expected to: 1. describe and illustrate the shape, properties, and characteristics of a normal distribution curve. 2. apply the z-score formula to convert a normal random variable to a standard normal variable and vice versa. 3. determine probabilities and percentiles using the standard normal distribution table (z-table) or appropriate technologies (e.g., calculator, software). 4. solve and interpret real-life problems involving normal distribution.																															
V. PROCEDURES	ACTIVITIES	ANNOTATION																														
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge																															
	Activity A.1: Leveling Learners Readiness	Activity A.1: Leveling Learners Readiness To determine learners’ prior knowledge of the lesson, you may facilitate any of the suggested activities.																														

Option 1
Self-Rating Scale of Understanding on Normal Distribution

Instructions:

1. Read each statement below.

Statement	Rating (1–4)
1. I understand what a random variable is.	
2. I can determine the probability distribution of a discrete random variable.	
3. I can interpret graphs of discrete probability distributions (bar graph/histogram).	
4. I understand the idea of a continuous random variable.	
5. I know what a bell-shaped distribution looks like.	
6. I can describe what mean and standard deviation tell about a set of data.	
7. I have encountered the term normal distribution before.	

2. Rate your level of understanding using this scale:

Rating

4 – I can explain this clearly to others.

3 – I understand this but need a bit more review.

2 – I have heard of this but I am not confident.

1 – This is new to me or I do not understand it.

3. Place your rating in the box beside each statement.

4. After rating, choose one item you rated 3 or 4 and write a short example or situation showing your understanding.

Option 1.
Self-Rating Scale of Understanding on Normal Distribution

This activity helps learners activate what they already know about data, variability, and probability distributions—ideas that form the foundation of the normal distribution—and by the end of the activity, learners should be able to reflect on their current understanding of key concepts related to data distribution, identify areas where they feel confident and areas that need clarification, and connect their prior knowledge of discrete and continuous random variables to the idea of a normal distribution.

Instructions:

Preparation

1. Prepare printed copies or display the table on the board/projector.
2. Ensure learners know the rating scale before starting.
3. Allot 10 minutes for rating and 5–7 minutes for sharing.

Guiding the Activity

1. Explain that the activity is not a test—it is for understanding learners' starting points.
2. Walk around while learners rate themselves; prompt those who seem unsure.
3. After learners complete the “example” and “question,” invite a few to share answers.

Bridging to the Lesson on Normal Distribution

Use their responses to connect prior concepts to the new lesson:

If many rated low on continuous random variables:

- Explain that the normal distribution is a continuous distribution used to model many real-life measurements.

	5. Choose one item you rated 1 or 2 and write a question you want clarified. Reflection Questions: 1. Which items did you rate the highest? What helped you learn these concepts before? 2. Which items received the lowest rating? Why do you think these topics are difficult for you? 3. How do you think these ideas relate to real-life situations involving measurement, scores, or natural variations? Option 2: Short, Targeted Quiz on Normal Distribution Instructions: Answer the following questions to the best of your ability. Each question can be multiple-choice, true/false, or a short calculation. Do not worry if you cannot answer all questions—this is to help you and the teacher understand what you already know. After completing the quiz, you will review your answers and note which items you feel confident about and which need clarification. 1. Multiple Choice Which of the following is a continuous random variable? a. Number of heads in 3-coin tosses b. Height of learners in your class	<i>If learners mentioned bell-shaped curves:</i> • Connect their mental image to the formal definition of normal distribution. <i>If learners show strong understanding of discrete distributions:</i> • Highlight how today they will shift from discrete to continuous probability distributions. Suggested Teacher Prompt “Based on your ratings, many of you already understand how data can vary and how probabilities work. Today, we will build on that knowledge by exploring the normal distribution, one of the most important and commonly used continuous probability distributions in real life.” Option 2: Short, Targeted Quiz on Normal Distribution This activity aims to assess learners’ existing knowledge of data, probability, and distributions in a focused and concise way, and by the end of the quiz, learners should be able to demonstrate their understanding of key concepts related to probability and data distribution, identify gaps in their knowledge before exploring the normal distribution, and connect their prior knowledge of discrete and continuous random variables to the concept of a normal distribution. Instructions: Preparation 1. Prepare printed quizzes or display on slides. 2. Decide whether the quiz is timed (5–10 minutes) or untimed. Facilitating the Activity 1. Give learners the quiz and allow them to work independently.
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	c. Number of books on a shelf d. Number of cars in a parking lot 2. True or False A probability distribution can be represented using a histogram. ☐ True ☐ False 3. Short Answer Write one example of a situation where data might form a bell-shaped distribution. 4. Which statement best describes the mean of a data set? a. The difference between the largest and smallest values b. The middle value when data are arranged in order c. The sum of all values divided by the number of values d. The value that appears most frequently 5. True or False The normal distribution is always symmetric around its mean. ☐ True ☐ False 6. Short Answer (Optional for advanced learners) What does the standard deviation tell you about a set of data?	2. Collect answers or discuss them immediately for a formative check. 3. Use learner responses to identify prior knowledge gaps and guide the introduction of the normal distribution. Facilitating the Reflection Questions 1. Which questions did you answer confidently? Why? 2. Which questions were difficult? What would you like to learn more about? 3. How do you think this connects to real-life data, like test scores, measurements, or natural phenomena? Bridging to the Lesson 1. Highlight items learners got correct, e.g., bell-shaped patterns, continuous variables. 2. Explain that the normal distribution is a special continuous distribution used to model many real-life phenomena.
A.2. Establishing the Purpose of the Lesson		
	Activity A.1: Appreciating Lesson Relevance Option 1: Comparing Discrete and Continuous Data Instructions: 1. Work in groups of 4-5. 2. Solve the two problems together. 3. Answer the questions and discuss your observations. Problem 1: Rolling a Die	Activity A.1: Appreciating Lesson Relevance To help learners appreciate the lesson and participate in class discussions, they may engage in any of the provided activities. Activity A.1 Option 1: Comparing Discrete and Continuous Data In this activity, learners will review what they know about discrete random variables—including mean, variance, and standard deviation—by working with dice rolls and then

What to do:
Roll one die 20 times and write down each result.
Count how many times each number (1 to 6) appears.

Tasks:

1. Determine the random variables and the probability.
2. Construct probability distribution table and draw its graph.
3. Find the expected value, variance and standard deviation
4. Answer the questions:
 - Which number appeared most?
 - Is the graph symmetric?
 - How spread out are the results?

Problem 2: Heights of Learners

What to do:

Look at these heights in centimeters:

145, 150, 152, 155, 155, 157, 160, 162, 165, 165, 168,
170

Tasks:

1. Draw a dot plot or histogram.
2. Calculate the expected value, variance, and standard deviation.
3. Answer the questions:
 - Can you assign exact probabilities to each height like the die?
 - Does the discrete formula work exactly for these heights? Why or why not?
 - How does the graph look? Symmetric? Bell-shaped?

Reflection Questions:

1. How are the dice results and heights different in terms of the type of data they represent?

explore continuous data such as student heights to observe how these differ from discrete data. Through hands-on group work, they will apply their knowledge of discrete probability distributions to calculate mean, variance, and standard deviation, observe patterns in continuous data and attempt similar calculations, compare discrete and continuous data to recognize why the same probability formulas do not work for continuous data, and identify the bell-shaped pattern in continuous data, preparing them to understand the normal distribution in the next lesson.

Preparation

Prepare tables or handouts for learners to record data, calculate probabilities, and plot graphs.
Ensure learners understand mean, variance, and standard deviation for discrete random variables.
Arrange classroom space for group work.

Instructions:

Step 1. Engage Learners

Ask learners:

- “What happens when we roll a die many times?”
- “Which numbers appear most often?”
- Review **mean, variance, and standard deviation** for discrete random variables.
- Tell learners:
- “Today we will explore both discrete and continuous data to see how they differ and prepare for learning about the normal distribution.”

Teacher Prompt: Highlight that the first problem is familiar (discrete), while the second introduces continuous data.

Step 2. Group Activity

Present Problem 1.

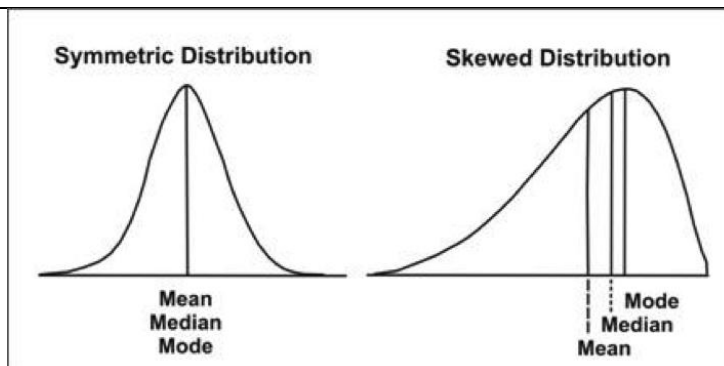
Teacher Prompts:

- “Which number appeared most?”

	- Why can't we use the same probability method for continuous data as we do for discrete outcomes? - What patterns do you notice in the histogram of heights or other continuous measurements? - Why do you think most real-life measurements tend to cluster around the average rather than the extremes? - Based on these observations, why is understanding the normal distribution important when interpreting or comparing real-life data?	"Does this match the mean?" "How spread out are the numbers?" Present Problem 2 Teacher Prompts: "Can we assign exact probabilities like the dice?" "How does the graph look? Symmetric? Bell-shaped?" Step 3: Group Discussion Ask learners: "How are dice results and heights different?" "Why can't we use the same probability method for continuous data?" "What pattern do you notice in the heights graph?" Step 4. Bridging to Normal Distribution Explain: "The histogram of heights is symmetric and bell-shaped. This is similar to a normal distribution." "Like discrete data, we can describe it using mean and standard deviation, but probabilities are calculated differently because the data is continuous." Highlight the difference - Discrete = specific outcomes, probability per outcome - Continuous = any value in a range, probability over intervals Teacher Prompt "You have now seen continuous data and patterns that lead to the normal distribution. Next lesson, we will formally learn how to use it." Notes for Teachers
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		• Encourage learners to verbalize observations during group work. • Compare the dice and heights graphs visually. • Focus on bridging prior knowledge to the concept of the normal distribution. • Present the learning goals and the performance task at the end of the unit.
B. Instituting New Knowledge	B.1. Presenting Examples	
	Activity B.1: Exploring Key Concepts Watch — Digest — Summarize Instructions: 1. Watch the video to be presented by the teacher. 2. As you watch, take quick notes: • What is a normal distribution (shape, mean, standard deviation)? • What does “standardizing” to a Z-score mean? • How do you use a Z-table (or calculator) to find probabilities or percentiles? • Did the video show any real-life data examples (e.g. heights, test scores)? 3. After watching, work individually or in pairs to write a short summary (~5 lines) answering: • What did you understand about normal and standard normal distributions? • When and why do we convert raw scores to Z-scores? • How can probabilities or percentiles help in real-life decisions (give one example, e.g. exam scores, health data)? 4. Answer one “thinking question”: “If I get a Z-score of +1.5, what does that tell me about where I stand compared to an average person in a normally distributed group?”	Activity B.1: Activity B.1: Exploring Key Concepts Watch — Digest — Summarize Instructions: 1. Select and queue up one of the suggested videos. Inform learners: <i>“You will watch a short video, but your goal isn’t just to ‘watch’ — it’s to understand and then explain in your own words. This helps you see how normal distribution works and how we use Z-scores in real situations.”</i> Suggested videos https://youtu.be/rzFX5NWojp0 https://youtu.be/2tuBREK_mgE https://youtu.be/CjF_yQ2N638 2. Play the video and ask learners to take notes focusing on key concepts listed above. 3. After the video, give learners time to write their summary (individually or in pairs). 4. Ask 2–3 volunteers to share their summaries or interesting ideas. Use this to emphasise: what a normal distribution describes, what Z-scores represent (above/below average), how to interpret probabilities/percentiles — linking to real-life context (grades, health measures, etc.). 5. Wrap up with a brief explanation: <i>“This activity shows how normal distribution and Z-scores allow us to standardize data, compare across different contexts, and compute probabilities. Later we’ll practice more</i>

		<i>problems, so you get comfortable with conversions and table lookup.”</i>
B.2. Discussing the Concept		
	Activity B.2: Deepening Conceptual Understanding Understanding Normal Random Variables and the Standard Normal Distribution Instructions: - Carefully read the lesson notes on sampling. - Take note of important terms. - Work through the step-by-step examples provided - Be ready to answer questions or share your thoughts during discussions. - Raise questions if their concepts are confusing. The Normal Distribution The normal distribution, also called the Gaussian distribution, is one of the most important probability distributions in mathematics and statistics. It was formally developed in the 18th–19th century by mathematicians Abraham de Moivre and later Carl Friedrich Gauss while studying natural errors in measurements. It is a continuous probability distribution that is symmetric about the mean (μ). It is known for its bell-shaped, symmetric curve, where: - Most values cluster around the mean (average), - The probabilities decrease evenly as values move away from the mean, - The mean, median, and mode are equal.	Activity B.2: Deepening Conceptual Understanding Understanding Normal Random Variables and the Standard Normal Distribution This lesson introduces learners to the concept of normal random variables and the normal distribution, including characteristics, probability calculations, and the use of the standard normal distribution (Z-values). Learners will see how normal distributions occur in real-life situations such as test scores, heights, and salaries. By the end of the lesson, learners will be able to interpret, sketch, and calculate probabilities under the normal curve and apply the Empirical Rule Prior to the discussion be reminded on the following: - Given the number of concepts to be covered, carefully plan your instructional approach to ensure learners can engage with the lesson without feeling overwhelmed. - Use visual aids like graphs and shaded areas to strengthen understanding. - Encourage learners to verbalize reasoning and explain steps to peers. - Integrate technology where possible. - Continuously relate abstract concepts to tangible, familiar examples. - The strategies provided are merely suggestions; you may apply the approach you believe best suits your learners during the discussion. - Ask check-in questions every concept and reflection question every end of the lesson.



The normal distribution appears naturally in many real-life situations such as:

- Heights and weights of people
- Test scores
- Measurement errors
- Natural and social phenomena

Because many random processes tend to form this pattern (especially with large data sets), the normal distribution became a foundation of probability theory, statistics, and data analysis, and it is widely used in education, science, economics, and quality control. It means that the normal distribution helps us describe, analyze, and predict patterns in data that show natural variation.

Characteristics of a Normal Distribution

A normal random variable has the following key characteristics:

1. Bell-shaped and Symmetric

- The graph looks like a bell.
- The left side is a mirror image of the right side it implies that the values are equally likely on both sides of the average.

2. Mean = Median = Mode

Strategy 1. Direct Instruction

Instructions:

1. Present the concept of normal random variables using slides or a whiteboard.
2. Explain the characteristics of a normal distribution: bell-shaped curve, symmetry, mean = median = mode, unimodal, asymptotic behavior.
3. Show examples from real life: test scores, heights, salaries.
4. Demonstrate step-by-step how to convert X-values to Z-values and read the Z-table.

Strategy 2. Guided Discovery

Instructions:

1. Divide learners into small groups.
2. Provide each group with a dataset (e.g., test scores, heights, or salaries).
3. Ask groups to:
4. Draw a rough normal curve for the data.
5. Identify the mean and standard deviation.
6. Calculate probabilities for certain ranges using Z-values.
7. Have each group present their results and reasoning to the class.

Strategy 3. Real-Life Application

Instructions:

1. Present a scenario (e.g., company salaries, student test scores).
2. Ask learners to:
3. Determine the mean and standard deviation from the data provided.
4. Convert X-values to Z-values and find probabilities using the Z-table.
5. Interpret the results in the context of the scenario.

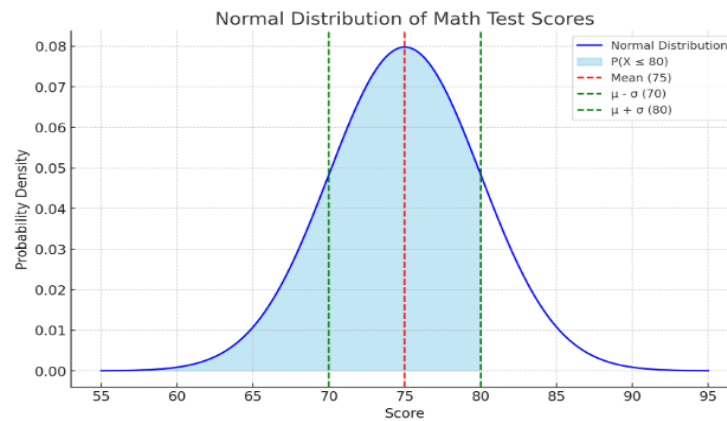
	- The mean (average), median, and mode are all at the center of the distribution it means that the most common value and the average are the same. 3. Single Peak (Unimodal) - The curve has one highest point it follows that there is one most frequent or typical value. 4. Asymptotic to the Horizontal Axis - The curve never touches the x-axis. - It extends infinitely in both directions meaning to say the extreme values are possible but very unlikely. 5. Defined by Two Parameters A normal random variable is described using: - Mean (μ) – the center of the distribution - Standard Deviation (σ) – the spread or variability Area Under the Normal Curve - The area under the curve represents the probability that a random variable (X) falls within a certain range. - The total area is probability of all outcomes is equal to one (1). - The mean divides the normal distribution curve into two equal parts. - The area between two points is probability that X falls in that interval. Steps to Sketch the Area Under the Normal Curve Draw the normal curve: - Draw a smooth, bell-shaped curve symmetric about the mean (μ). - Mark the mean μ on the horizontal axis. Label key points:	6. Facilitate a class discussion on implications, trends, and decision-making based on probabilities. Reflection Questions - What does it mean when a distribution is symmetric? - How do you find the probability of a score being above the mean? - What is the significance of converting X to Z? - What does the shaded area under your curve represent? - Were there any surprises in the probabilities you calculated - How can this probability information help decision-making? - How does the concept of normal distribution help in understanding real-life patterns?
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- Mark $\mu \pm \sigma$ (one standard deviation away from the mean).

3. Shade the area of interest:

- Identify the range for which you want the probability.
- Shade the region under the curve corresponding to this range.

4. Interpret the shaded area:



- The shaded area represents the probability of the random variable falling within that range.

Example 1.

Suppose the math test scores of a large group of students are normally distributed.

- Mean score (μ) = 75
- Standard deviation (σ) = 5

Draw the normal distribution showing that the learners score lower than 80.

The illustration above is the normal distribution curve for the math test scores:

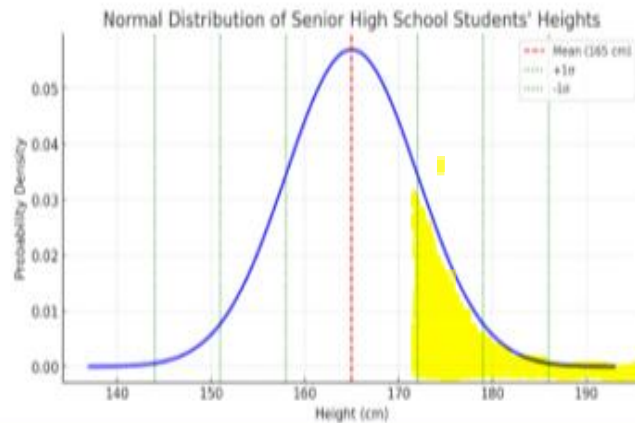
- The mean (75) is marked with a red dashed line at the center.
- One standard deviation from the mean (70 and 80) is marked with green dashed lines.
- The shaded area represents $P(X \leq 80)$, the probability of a student scoring 80 or less (~84.13%).

Example 2.

Let the heights of senior high school students be normally distributed.

- Mean height = 165 cm
- Standard deviation = 7 cm

Draw the normal distribution showing that the learners height is greater than 170 cm



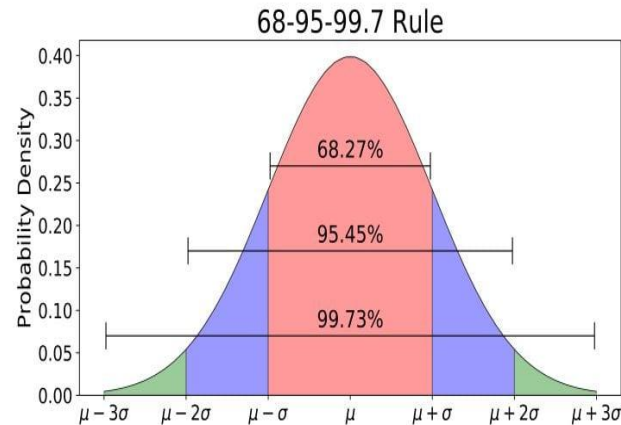
The illustration above is the normal distribution curve for the heights of senior high school students:

- Mean (μ): 165 cm (red dashed line at the peak)
- Standard deviations (σ): 7 cm, shown as green dotted lines at $\pm 1\sigma$, $\pm 2\sigma$, $\pm 3\sigma$
- The curve is symmetric, with most students' heights concentrated near the mean.

- The shaded area represents $P(X \geq 170)$, the probability of a student scoring 80 or less (~84.13%).

The Empirical Rule (68–95–99.7 Rule)

This rule helps describe how values are spread in a normal random variable.



Using the test score in example 1

- About **68%** of students scored between **70 and 80** ($\mu \pm 1\sigma$)
- About **95%** scored between **65 and 85** ($\mu \pm 2\sigma$)
- About **99.7%** scored between **60 and 90** ($\mu \pm 3\sigma$)

The Standard Normal Distribution

A **normal random variable** X has:

- Mean: μ
- Standard deviation: σ

Its probability distribution is called a normal distribution.

A **standard normal distribution denoted by Z** is a special case of the normal distribution with:

- Mean: 0
- Standard deviation: 1

Why convert?

Converting X to Z allows us to use standard normal tables (Z-tables) to find probabilities easily.

Formula for Conversion**Converting X to Z**

$$Z = \frac{X - \mu}{\sigma}$$

- X = value of the normal random variable
- μ = mean
- σ = standard deviation
- Z = corresponding standard normal value

Interpretation:

Z tells how many **standard deviations X is away from the mean.**

Converting Z to X

$$X = Z \cdot \sigma + \mu$$

- Z = standard normal value
- μ = mean
- σ = standard deviation
- X = value of the original normal variable

Example 3. Converting X to Z

The heights of students are normally distributed with $\mu = 165$ cm, $\sigma = 7$ cm.

Find the standard normal variable Z corresponding to a student with **height 172 cm.**

Solutions:

$$Z = \frac{X - \mu}{\sigma} = \frac{172 - 165}{7} = \frac{7}{7} = 1$$

Interpretation:

The height 172 cm is 1 standard deviation above the mean.

Example 4. Converting Z to X

Find the height of a student corresponding to $Z = -1.5$.

Solutions:

$$X = Z \cdot \sigma + \mu = (-1.5)(7) + 165 = -10.5 + 165 = 154.5 \text{ cm}$$

Interpretation:

A student with $Z = -1.5$ has a height 1.5 standard deviations below the mean, which is 154.5 cm.

The Z-Table

A **Z-table** (or standard normal table) shows the area (probability) under the standard normal distribution curve to the left of a given Z-value.

Recall:

- A Z-value tells how many standard deviations a value is from the mean.
- The standard normal distribution has:
 - Mean = 0
 - Standard deviation = 1
- The total area under the curve = 1.00 (100%)

The Z-table helps us answer probability questions once we convert a normal random variable X into a standard normal variable Z .

Structure of the Z-Table

The Z-table is usually arranged as follows:

- **Left column:** the whole number and first decimal of the Z-value
- **Top row:** the second decimal of the Z-value
- **Body of the table:** the area to the left of that Z-value

How to Read the Z-Table

Example 5.

Find the value of $Z=1.23$.

Step 1. **Identify the Z-value**

$$Z = 1.23$$

Step 2. **Locate the row**

$$\text{Row} = 1.2$$

Step 3. **Locate the column**

$$\text{Column} = 0.03$$

Step 4. **Read the value**

$$\text{Table value} \approx \mathbf{0.8907}$$

Interpretation:

$$P(Z \leq 1.23) = 0.8907$$

This means **89.07%** of the values lie **below** $Z = 1.23$.

Understanding Areas Using the Z-Table

A. Probability to the Left

Directly read from the Z-table.

Example 5:

$$P(Z \leq 0.50) = 0.6915$$

B. Probability to the Right

Since total area = 1:

$$P(Z > a) = 1 - P(Z \leq a)$$

Example 6:

$$P(Z > 1.25) = 1 - 0.8944 = 0.1056$$

C. Probability Between Two Z-Values

$$P(a < Z < b) = P(Z \leq b) - P(Z \leq a)$$

Example. 7

Find $P(0.50 < Z < 1.50)$

From the table:

- $P(Z \leq 1.50) = 0.9332$

- $P(Z \leq 0.50) = 0.6915$

$$P = 0.9332 - 0.6915 = 0.2417$$

About 24.17% of the values lie between $Z = 0.50$ and $Z = 1.50$.

Z-Table with Negative Z-Values

The standard normal curve is **symmetric** about 0.

Key idea:

$$P(Z \leq -a) = 1 - P(Z \leq a)$$

Example 8.

Find $P(Z \leq -1.20)$

From table:

$$P(Z \leq 1.20) = 0.8849$$

$$P(Z \leq -1.20) = 1 - 0.8849 = 0.1151$$

About **11.51%** lie below $Z = -1.20$.

Word Problems**Example 9.**

The heights of learners are normally distributed with $\mu=165$ "cm", $\sigma=7$ "cm". What is the probability that a learner is taller than 172 cm?

Step 1. **Convert to Z**

$$Z = \frac{172 - 165}{7} = 1$$

Step 2. **Use the Z-table**

$$P(Z \leq 1) = 0.8413$$

Step 3. **Probability taller**

$$P(X > 172) = 1 - 0.8413 = 0.1587 \approx 16\%$$

Example 10.

Test scores are normally distributed with $\mu = 75$, $\sigma = 5$.
Find the probability that a student scores **above 80**.

Step 1. **Convert X to Z**

$$Z = \frac{80 - 75}{5} = 1$$

Step 2. **Use the Z-table**

$$P(Z \leq 1.00) = 0.8413$$

Step 3. **Find probability above**

$$P(X > 80) = 1 - 0.8413 = 0.1587$$

About **15.87%** of students scored above 80.

Example 11.

Suppose the math test scores of a large group of senior high school students are normally distributed.

Mean score (μ) = 75

Standard deviation (σ) = 5

The school wants to know the proportion of students whose scores are neither too low nor too high, specifically those who scored between 72.5 and 81 points.

Solutions:

Step 1: **Convert the scores to Z-values**

For **72.5**:

$$Z = \frac{72.5 - 75}{5} = -0.5$$

For **81**:

$$Z = \frac{81 - 75}{5} = 1.2$$

Step 2: **State the probability in terms of Z**

$$P(72.5 \leq X \leq 81) = P(-0.5 \leq Z \leq 1.2)$$

Step 3: Use the Z-table

From the Z-table:

$$P(Z \leq 1.2) = 0.8849$$

$$P(Z \leq -0.5) = 0.3085$$

Step 4: Compute the probability

$$\begin{aligned} P(-0.5 \leq Z \leq 1.2) &= 0.8849 - 0.3085 \\ &= 0.5764 \end{aligned}$$

Interpretation

About 57.64% of the students scored between 72.5 and 81 on the math test. This means more than half of the students fall within this “average to above-average” score range.

Example 12.

Suppose the monthly salaries of employees in a company are normally distributed.

- Mean monthly salary (μ) = ₱30,000
- Standard deviation (σ) = ₱4,000

The management wants to know the percentage of employees who earn a moderate income, neither very low nor very high. Specifically, they want to find the proportion of employees earning between ₱28,000 and ₱34,800 per month.

Solutions:

Step 1. Convert the salaries to Z-values

For **₱28,000**:

$$Z = \frac{28,000 - 30,000}{4,000} = -0.5$$

For **₱34,800**:

$$Z = \frac{34,800 - 30,000}{4,000} = 1.2$$

	Step 2. Express the probability in terms of Z $P(28,000 \leq X \leq 34,800) = P(-0.5 \leq Z \leq 1.2)$ Step 3. Use the Z-table From the Z-table: $P(Z \leq 1.2) = 0.8849$ $P(Z \leq -0.5) = 0.3085$ Step 4. Compute the probability $P(-0.5 \leq Z \leq 1.2) = 0.8849 - 0.3085$ $= 0.5764$ Interpretation About 57.64% of the employees earn between ₱28,000 and ₱34,800 per month. This means more than half of the company's employees fall within this salary range. By completing these activities, you now understand how normal random variables help describe real-world data, how to use the standard normal distribution to calculate probabilities, and how to interpret results. Remember, these concepts are not just theoretical—they are powerful tools for analyzing patterns in education, science, business, and daily life.	
B.3. Developing Mastery		
	Activity 1: Practicing Learned Skills Instructions: Do as indicated. Write your answer on a separate sheet of paper. Part A. 1. On graph paper, sketch a bell curve for test scores with mean = 50 and standard deviation = 10. 2. Mark the intervals: 40–60, 30–70, and 20–80. 3. Shade each region and label the approximate % of data it covers.	Activity 1: Practicing Learned Skills This activity helps learners understand the shape and symmetry of the normal curve, recognize how data cluster around the mean, and see how standard deviations represent spread. It strengthens skills in computing and interpreting z-scores, shows how different datasets can be compared using a common scale, and builds understanding of probabilities and areas under the curve. Through real-life examples, learners practice using z-tables or technology to interpret results, relate probabilities to actual situations,

	Part B. 1. Compute z for these cases: a. $X = 70, \mu = 60, \sigma = 5$ b. $X = 45, \mu = 50, \sigma = 10$ 2. Convert back to raw scores X given: a. $z = 2, \mu = 100, \sigma = 15$ b. $z = -1.2, \mu = 80, \sigma = 8$ Part C. Read the given scenario: IQ scores are normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 15$. Find the following probabilities: a. $P(X < 115)$ b. $P(X > 85)$ c. $P(85 < X < 115)$ Part D. Read the given scenario: SAT scores are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Answer the following questions: a. What percentage of learners score above 650? b. What score marks the lowest 25% of learners? Before answering the main tasks, respond to these questions: 1. What formula is used to convert a raw score (X) into a z-score? 2. If $X = 650$, what z-score will this correspond to? 3. What does the z-score tell us about the position of 650 relative to the mean? 4. When solving for probabilities greater than a score (e.g., $P(X > 650)$), do we directly use the z-table value or subtract from 1? Why?	and appreciate the meaning and fairness of standardized scores. To facilitate the activity, you may follow of the suggested strategies. Strategy 1: Structured Independent Work with Checkpoints Instructions: Orientation (Before Work Begins): 1. Briefly explain the goal of Activity 1 and how Parts A–D are connected (graphing $\rightarrow$ computation $\rightarrow$ probability $\rightarrow$ interpretation). 2. Review only the <i>key formulas</i> (z-score formula, converting Z back to X) without solving examples. Independent Work with Checkpoints: 1. Learners work individually on the activity. 2. The teacher announces <i>checkpoints</i> after each part: 3. After Part A: “Check that your curve is symmetrical and your shaded areas align with the Empirical Rule.” 4. After Part B: “Verify if your z-values are positive or negative and reflect position relative to the mean.” 5. Address common errors noticed while circulating. Quick Clarification Bursts: 1. Pause briefly when many learners show the same difficulty (e.g., using the Z-table or interpreting percentiles). 2. Provide quick reminders instead of full re-teaching. Strategy 2. Think–Pair–Share with Visual Emphasis Instructions: 1. Think:
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5. What does the phrase “lowest 25% of learners” mean in terms of area under the normal curve?
6. From the z-table, what approximate z-score corresponds to the 25th percentile?
7. How do you convert a z-score back into a raw score (X)?

- Ask learners to answer the *pre-activity Reflection Question* (z-score formula, interpretation, percentiles) individually.
- Give 2–3 minutes of quiet thinking time.

2. **Pair:**

- Learners pair up to compare answers and reasoning.
- Assign each pair to focus on one part of the activity (e.g., one pair explains Part A shading, another explains Z-table use).

3. **Share:**

- Selected pairs share their answers on the board or orally.

Answer Key:

Part A.

- 40–60 → ~68% (approximately 68%)
- 30–70 → ~95% (approximately 95%)
- 20–80 → ~99.7% (approximately 99.7%)

Part B.

$$1a. z = \frac{70 - 60}{5} = 2.0$$

$$1b. z = \frac{45 - 50}{10} = -0.5$$

$$2a. X = 100 + (2)(15) = 130$$

$$2b. X = 80 + (-1.2)(8) = 70.4$$

Part C.

$$a. z = 1.0 \rightarrow P = 0.8413$$

$$b. z = -1.0 \rightarrow P = 0.8413$$

$$c. \text{Between } -1 \text{ and } 1 \rightarrow P = 0.6826$$

Part D.

$$1. z = \frac{X - \mu}{\sigma}$$

$$2. z = \frac{650 - 500}{100} = 1.5$$

3. The score 650 is 1.5 standard deviations above the mean.

		4. We subtract from 1 because the z-table gives the area to the left of z. For “greater than,” we It means we are looking for the 25th percentile, or the score where 25% of the distribution lies below it. 5. For area = 0.25, $z \approx -0.67$. 6. To convert a z-score back into a raw score, use the formula: $X = \mu + (z \cdot \sigma)$
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Making Real-World Connection Instructions Part A. Data Collection - Conduct a quick survey using the question: “How many hours do you spend on screen (phone, tablet, laptop) per day?” - Select at least of your classmates or nearby sections. - Record the data in a table. Use whole numbers or nearest 0.5 hour for accuracy. Part B. Data Analysis - Compute the: - Mean (μ) - Standard deviation (σ) - Make an assumption: <i>“We assume that the daily screen time of students follows a normal distribution.”</i> Part C. Application of Normal Distribution - Conversion to Standard Normal Variable Choose one data value from your survey and compute its Z-score using $Z = \frac{X - \mu}{\sigma}$ - Probabilities and Percentiles Using a Z-table or calculator/app:	Activity C.1: Making Real-World Connection Learners will collect real data on daily screen time, analyze it using the normal distribution, and make recommendations based on probabilities and percentiles. Instructions: - Give time to learners for data gathering - Allow the use of phones/calculators for computation - Remind learners that the accuracy of interpretation matters more than “perfect” data. - Emphasize that this task is an application, not mastery. - Facilitate the presentation. - Synthesize the activity by asking the following reflective questions. - What did you learn about normal distribution after collecting and analyzing real data from your classmates? - How did converting raw data to Z-scores and using probabilities or percentiles help you better understand the spread and pattern of the data? - Why is normal distribution important in making real-life decisions, such as identifying students who may need support or interventions? - What part of today’s activity helped you appreciate statistics more, and why? Rubric:

	a. Find the probability that a randomly selected student spends more than X hours on screen (choose X from your data).	Criteria	Points
	b. Determine the 75th percentile of screen time.		
	3. Problem Solving & Decision-Making		
	Suppose the school plans a digital wellness program for students in the top 25% of screen time users. Determine the screen-time cutoff value. Explain how you obtained this value.		
Part D. Presentation of Findings			
1. Draw a normal distribution curve.		Data collection & sampling	2
2. Label the mean and standard deviation		Correct μ and σ	2
3. Shade the region representing the top 25%		Z-score & probability computations	3
4. Write a 2–3 sentence conclusion explaining how the results can guide school decisions.		Decision-making & justification	2
		Graph & clarity of conclusion	1
C.2. Making Generalization			
Activity C.2: Wrapping Up the Lesson		Activity C.2: Making Real-World Connection	
Option 1. Exit Ticket		These wrap-up activities help learners consolidate their understanding of normal distributions and standard normal distributions. The Exit Ticket encourages individual reflection and personal connection to real-life examples, while Expert Talks provide opportunities for learners to share insights, observe peer understanding, and deepen conceptual awareness. Together, these activities reinforce comprehension, encourage critical thinking, and connect theoretical concepts to everyday situations.	
Instructions:		Option 1. Exit Ticket	
1. Think of a real-life situation where a normal distribution occurs (e.g., test scores, employee salaries, heights).		Instructions:	
2. Write down:		1. Explain the purpose of the exit ticket: learners will reflect on their understanding of normal distribution and share one real-life example.	
• Your example		2. Ask learners to write down:	
• One question you still have about the lesson		• A real-life example of normal distribution	
• One thing you learned today that you found interesting or surprising			
3. Submit your responses to the teacher before leaving the class.			

	Option 2. Expert Talks – Sharing Observations Instructions: 1. Listen carefully as your classmates (experts) share their observations about normal distributions in real life. 2. Take notes on examples, interpretations, and interesting points shared by your peers. 3. Be ready to ask one thoughtful question or share one observation of your own after each talk.	• One question they still have about the lesson • One key takeaway or interesting point from the lesson 3. Collect the exit tickets before the learners leave the class. 4. Quickly review the responses to identify common misconceptions or areas that may need reinforcement in future lessons. Option 2. Expert Talks – Sharing Observations Instructions: 1. Select a few learners to serve as “experts” who will share observations or examples of normal distribution in real life. 2. Give each expert 2–3 minutes to present their observation, explanation, or real-life example. 3. Encourage the rest of the class to actively listen, take notes, and prepare at least one question or observation to share after each talk. 4. Facilitate a short discussion after each expert talk to clarify ideas, highlight key points, and link observations to lesson objectives. 5. Conclude by summarizing the main takeaways from the talks, reinforcing the concepts of mean, standard deviation, Z-values, and interpretation of probabilities.
C.3. Evaluating Learning		
	Activity C.3: Assessing Learning Outcomes Instructions Read each scenario carefully and perform the task as directed. 1. A test has $\mu = 50$ and $\sigma = 8$. If a learner scores 66, what is the z-score? 2. If $z = -1.25$, $\mu = 100$, $\sigma = 12$, what is the raw score?	Activity C.3: Assessing Learning Outcomes This activity is designed as independent work to assess learners’ proficiency in applying normal distribution concepts. Learners will practice converting raw scores to Z-scores and vice versa, calculating probabilities using Z-tables, and interpreting results in real-life contexts. This activity allows the teacher to evaluate each learner’s understanding and ability to apply the lesson independently.

3. The scores on a standardized test are normally distributed with a mean of 500 and a standard deviation of 90. Determine:
 - a. $P(X < 560)$
 - b. $P(X > 440)$
4. The weights of bags of rice are normally distributed with $\mu = 50$ kg and $\sigma = 2$ kg.
 - a. What % of the bags weigh more than 53 kg?
 - b. A retailer only accepts bags in the middle 80% of weights. What is the acceptable weight range?

Reflection Questions

1. Which question did you find the most challenging, and why?
2. How did calculating Z-scores or probabilities help you understand the scenario better?
3. Can you think of a real-life situation where you might use what you learned in this activity?

Instructions:

1. Explain that learners will independently solve problems using normal distribution concepts, including Z-scores and probabilities. Emphasize careful reading of each scenario.
2. Provide learners with the worksheet or display the scenarios on the board/projector.
3. Remind learners that this is independent work meant to assess their proficiency. Encourage them to show all steps clearly.
4. Walk around the class to observe progress, answer clarification questions if needed, but avoid giving direct solutions.
5. After learners complete the tasks, review the answers step by step. Use graphs or sketches of the normal curve to visually explain probabilities, emphasizing interpretation in context.
6. Highlight common mistakes, reinforce correct methods, and discuss practical applications such as test scores, product weights, or quality control.

Answer Key:

1. $z = \frac{66 - 50}{8} = 2.0.$
2. $X = 100 + (-1.25)(12) = 100 - 15 = 85$
3. a. $z = \frac{560 - 500}{90} = 0.67 \rightarrow P(X < 560) = 0.7486.$
 b. $z = \frac{440 - 500}{90} = -0.67 \rightarrow P(X > 440) = 1 - 0.2514 = 0.7486.$
4. a. $z = \frac{53 - 50}{2} = 1.5.$
 $P(X > 53) = 1 - 0.9332 = 0.0668 \rightarrow 6.7\%.$
 b. Middle 80% $\rightarrow$ cutoff = 10% in each tail.
 For 10% left area, $z \approx -1.28;$
 For 90% left area, $z \approx +1.28.$
 $X = 50 + (-1.28)(2) = 47.44.$

		$X = 50 + (1.28)(2) = 52.56.$ Acceptable range = 47.44 kg to 52.56 kg.
C.4. Additional Activities		
	Activity C.4: Extending and Reinforcing Learning Instructions Read each scenario carefully and perform the task as directed. Remedial Activities: For each question, follow the steps: - Write the formula for z: $z = \frac{X - \mu}{\sigma}$. - Substitute the values. - Compute step by step. - Find the missing value. - If $X = 60$, $\mu = 50$, $\sigma = 5$, find z. - If $z = -2$, $\mu = 100$, $\sigma = 15$, find X. - The IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. - Find: $P(X < 115)$. - Find $P(X > 85)$. - The lifespan of batteries is normally distributed with $\mu = 200$ hours and $\sigma = 20$ hours. Answer the following questions: - What % of batteries last more than 220 hours? - What is the cutoff lifespan for the bottom 5% of batteries? Enrichment Activities: - Draw two normal curves on the same graph: - Curve A: $\mu = 170$ cm, $\sigma = 5$ cm (learners' heights in School A). - Curve B: $\mu = 170$ cm, $\sigma = 10$ cm (learners' heights in School B). - Label both means.	Activity C.4: Extending and Reinforcing Learning These activities provide learners with opportunities to strengthen and extend their understanding of normal distributions, Z-scores, and probability concepts. Remedial Activities are designed to reinforce foundational skills for learners who need additional practice with Z-score calculations, probability interpretation, and identifying values from a normal distribution. Enrichment Activities challenge learners to apply their knowledge to more complex, comparative, and real-life scenarios, including graph analysis, interpreting variability, and problem-solving in practical contexts such as test scores, heights, and scholarship cutoffs. Together, these activities support differentiated instruction by addressing varying learner proficiency levels and promoting conceptual understanding, application, and critical thinking. Instructions: Introduce the Purpose: - Explain that the activities are differentiated to address varying proficiency levels. - Emphasize that learners will practice Z-score calculations, probability determinations, and interpretation of normal distributions in real-life contexts. Distribute Materials: - Provide learners with the appropriate set of activities (Remedial or Enrichment), or display on the board/projector.

	c. Compare: Which curve is narrower? Which is wider? What does this tell you about variation in heights? 2. Compare the performance of the two learners relative to their respective groups. Who performed better, and why? Consider the mean, standard deviation, and each learner's score in your explanation <i>Alice: Test A $\rightarrow \mu = 70, \sigma = 8, \text{score} = 82.$</i> <i>Ben: Test B $\rightarrow \mu = 500, \sigma = 100, \text{score} = 630.$</i> 3. Analyze the situation below and answer the questions that follow. The SAT scores are normally distributed with a mean of 500 and a standard deviation of 90. a. The top 5% of learners receive a scholarship. What is the minimum SAT score needed? b. What % of learners score between 450 and 600?	3. Clarify Steps and Expectations: For Remedial Activities: - Remind learners to write the Z-score formula $z = \frac{x - \mu}{\sigma}$, substitute values, and compute step by step. - Solve for missing values, calculate probabilities using Z-tables, and interpret results in context. For Enrichment Activities: - Encourage learners to draw and label normal curves, compare distributions, interpret variation, and evaluate relative performance. - Solve real-life problems (e.g., test scores, scholarship cutoffs) using Z-scores and probability concepts. 4. Monitor Learners: - Walk around the classroom to observe, provide guidance, and clarify misconceptions without giving direct solutions. - Encourage peer discussion and reasoning for enrichment tasks. 5. Review and Discuss: - Go over solutions for both remedial and enrichment tasks. - Use graphs and sketches to visually demonstrate normal distributions and shaded probability areas. - Highlight key concepts: mean, standard deviation, Z-scores, probability interpretation, and real-life applications. 6. Provide Feedback: - Emphasize correct methods and interpretation. - Address common errors or misconceptions. - Encourage reflection on how these statistical concepts can be applied in everyday situations.
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Answer Key:**Remedial Activities:**

1. a. $z = \frac{60 - 50}{5} = \frac{10}{5} = 2.$

b. $X = 100 + (-2)(15) = 100 - 30 = 70.$

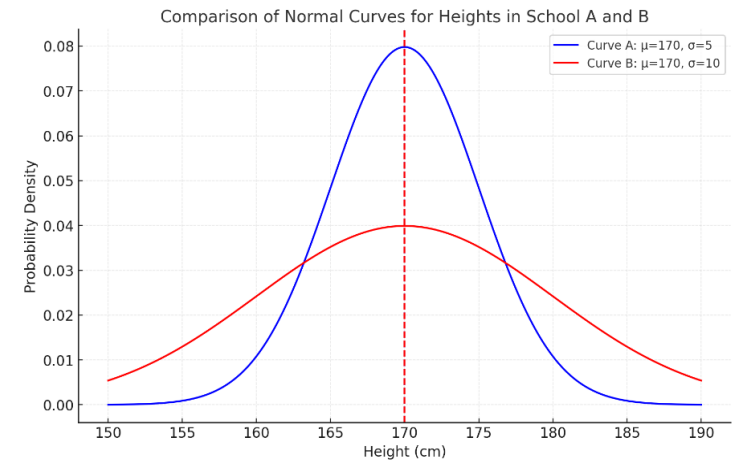
2. a. $z = \frac{115 - 100}{15} = 1.0 \rightarrow \text{area} = 0.8413 \rightarrow P = 0.8413.$

b. $z = \frac{85 - 100}{15} = -1.0 \rightarrow \text{area} = 0.1587 \rightarrow P(X > 85) = 1 - 0.1587 = 0.8413.$

3. a. $z = \frac{220 - 200}{20} = 1.0. P(X > 220) = 1 - 0.8413 = 0.1587 = 15.9\%.$

b. Bottom 5% $\rightarrow z \approx -1.645.$

$X = 200 + (-1.645)(20) = 200 - 32.9 = 167.1 \text{ hours}.$

Enrichment Activities:

- Curve A (blue) is narrower $\rightarrow$ smaller spread in School A's heights.
- Curve B (red) is wider $\rightarrow$ more variability in School B's heights.
- Interpretation: Curve A is narrower, meaning learners' heights in School A are more consistent

		(less variation). Curve B is wider, meaning there's more diversity in learners' heights in School B. 2. Alice: $z = \frac{82 - 70}{\frac{8}{\sqrt{100}}} = \frac{12}{8} = 1.5$. Ben: $z = \frac{630 - 500}{\frac{130}{\sqrt{100}}} = \frac{130}{100} = 1.3$. Alice did slightly better relative to her peers since $1.5 > 1.3$. 3. a. Top 5% $\rightarrow z = 1.645$. $X = 500 + (1.645)(90) = 500 + 148.05 = 648.05$. Cutoff ≈ 648. b. $z(450) = \frac{450 - 500}{\frac{90}{\sqrt{100}}} = -0.56 \rightarrow \text{area} = 0.2877$. $z(600) = \frac{600 - 500}{\frac{90}{\sqrt{100}}} = 1.11 \rightarrow \text{area} = 0.8665$. Probability $= 0.8665 - 0.2877 = 0.5788 = 57.9\%$.
III. CONTENT	Lesson 9.3: Sampling (Suggested Time Allotment: 3 hours)	
IV. OBJECTIVES	At the end of the lesson, the learners are expected to - define random sampling and its types. - recommend appropriate sample size for a given investigation. - differentiate between a parameter and a statistic. - compare probability and nonprobability sampling techniques based on their characteristics and applications.	
V. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge	
	Activity A.1 Leveling Learner Readiness Option 1: Sampling Matters Scenario: Your school wants to find out what snacks learners prefer to sell in the canteen next semester. Due to time and resource limitations, they cannot survey all learners, so they plan to select a smaller group instead. Instructions: In your assigned group (3–5 members), discuss the possible challenges of surveying all learners.	Activity A.1 Leveling Learner Readiness This activity introduces the purpose of sampling through a relatable school scenario. It helps learners understand the reasons behind using samples instead of full populations and prepares them to explore key terms like random sampling, representativeness, and sample size. To facilitate this activity, the teacher may apply one of the following strategies: Strategy 1: Contextual Group Task: Sampling Matters Instructions:

2. List at least three reasons why surveying everyone may not be practical.
3. Propose a fair and logical way to select a smaller group of learners (a sample) whose preferences may represent the whole learner body.
4. Record your responses on a sheet of paper or digital slide.
5. Assign a representative to present your group's ideas to the class.

Reflection Question:

1. What issues might arise if only one section or grade level is surveyed?
2. How does random selection help improve the accuracy of the results?
3. Why is sampling a common method used in research?

Option 2: Voices in Focus

Instructions:

1. **Reflect quietly on the question:**
 - “Have you ever participated in a school survey or voted in a learner election?”
2. **Quick Poll:**
 - Participate in a brief survey conducted by the teacher. For example, the teacher may ask:
 - “Which school activity do you prefer: sports, arts, or science clubs?”

1. Prepare printed or digital copies of the scenario and guiding questions.
2. Group learners into teams of 3–5 to encourage collaborative discussion.
3. Introduce the activity by saying: “Today, you will take on the role of researchers helping the school canteen improve its menu. Let’s think about how we can make good decisions using data from a smaller group of learners.”
4. Allow 7–10 minutes for group discussion and preparation.
5. Invite selected groups to share their proposed sampling plans with the class.
6. Facilitate a short discussion focusing on feasibility, bias, time constraints, and the need for random selection in drawing valid conclusions

Possible Answer:

1.	Takes too much time; Costs more (money and staff); Hard to organize
2.	Too many learners; Needs more people; Hard to check all answers
3.	Pick learners randomly; Choose from each class or grade Include different groups (age, gender, etc.)

Strategy 2:

Digital or Physical Poll with Reflection -Voices in Focus

Instructions:

1. Conduct a short poll on a relevant, learner-friendly topic (e.g., favorite snack, preferred school event, etc.).
2. Display and discuss poll results to generate interest.
3. Guide the class through a reflective discussion based on their answers and the limitations of a small data set.
4. Introduce basic sampling concepts: population, sample, bias, and representativeness.

	- Respond by raising your hand or using a digital polling tool such as Mentimeter or Google Forms. This poll will provide a snapshot of learner preferences 3. Follow-Up Question: Consider the question: “Do you think your individual response, or the poll results, accurately represent the opinions of the entire school community? Why or why not?” 4. Guided Discussion: Participate in a teacher-led discussion to explore: - How surveys are conducted and their purpose. - Why sampling is often preferred over surveying every individual. - What makes a sample fair and representative (e.g., randomness, diversity). - Real-life examples connecting these concepts to your own experiences Option 3: Mini-Debate Instructions: - Divide the class into two teams: - Team A argues in favor of using a census (asking everyone). - Team B argues in favor of using sampling (asking only a selected group). - Each team is given 5 minutes to prepare their arguments. - Teams take turns presenting (2–3 minutes per team). - After both sides have spoken, reflect as a class: “Which method seems more practical and effective in most real-life situations?”	- Use this activity to set the stage for more in-depth study of sampling methods. Possible Answer: - Results won’t reflect the whole school. - It gives everyone a fair chance and avoids bias. - Saves time and money; Still gives useful results. - To collect opinions or information Understand what people think - Faster and cheaper Easier to manage - Random selection Includes different types of people - Election polls Customer feedback - School activity surveys Strategy 3: Structured Argumentation and Critical Thinking Instructions: - Divide the class into two teams. - Give learners 5 minutes to prepare arguments. - Each team presents their side in 2 minutes. - End with a short teacher-led discussion: “Sampling is used when we want results that are accurate, but also practical. In this lesson, you’ll learn how to choose a good sample and avoid common mistakes.”
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	A.2. Establishing the Purpose of the Lesson	
	Activity A.2: Appreciating Learning Relevance Option 1: Crash and Learn Scenario: A group of learners wants to know which school event learners like best. They only ask learners from the Academic Track. Everyone says they prefer the Business Fair. The group concludes it's the most loved event in the whole school. Instructions: 1. In pairs, analyze the scenario. 2. Discuss what went wrong in their sampling method 3. Suggest how to improve it. 4. Record your answers on a half-sheet of paper. 5. Be ready to share your response with the class. Reflection Questions 1. What's unfair about the sample? 2. Who was left out? 3. How could you make it more accurate? Option 2: The Sampling Dilemma	Activity A.2: Appreciating Learning Relevance This activity introduces the importance of proper sampling through engaging, real-life school scenarios. It helps learners understand how poor sampling choices can lead to biased or inaccurate conclusions. This sets the purpose of learning about sampling techniques in upcoming lessons. Strategy 1: Scenario Analysis – Crash and Learn Instructions: 1. Present the scenario either through printed handouts, slide presentation, or verbal storytelling. 2. Allow learners to reflect individually, then pair up to discuss. 3. Facilitate a guided class sharing to highlight key issues like sampling bias, exclusion, and lack of representativeness. 4. Link to real-world decision-making and school-wide planning. Note: The teacher may contextualize the scenario. Possible Answers for Reflection Questions Option 1: Crash and Learn 1. The sample is unfair; it only includes one strand. 2. Other strands and grades were left out. 3. Select learners randomly from all strands and grades. Strategy 2: Think–Pair–Share and Consensus Building Instructions:

	Scenario-Based Prompts: Imagine you are a researcher. Your class wants to know what school event to organize next. You have two options for collecting opinions: Prompt 1: Would you rather... 1. Ask 100 random learners from different grade levels and tracks? 2. Ask your 100 closest classmates from the same track? Prompt 2: Would you rather... 1. Ask only the most active learners who usually attend school events? 2. Include both active and non-active learners randomly? Instructions: 1. Think individually and write your answers for both prompts on a sheet or digital device. 2. Form a group of 4–5 members and share your answers. 3. As a group, agree on the most effective sampling method for each prompt and explain your reason. 4. Choose one group member to share your group’s insights with the class. Reflection Questions: 1. Which method leads to more reliable results? 2. What are the dangers of choosing people who are too similar to you? 3. How do randomness and diversity improve data collection? Option 3: Survey Fail Detective Scenario: You’re a survey detective! The principal gets three surveys from learners about school uniforms. Each group used a different method: Group A: Asked only Grade 11 boys	1. Begin by explaining the idea of “sampling dilemmas” in research – where researchers must choose the best sampling strategy for fairness and accuracy. 2. Present each prompt on the board or screen. 3. Let learners reflect and record their individual answers. 4. Encourage collaborative decision-making by forming small discussion groups. 5. Allow 5–7 minutes for group discussion and then hear selected group insights. 6. Emphasize concepts like sampling bias, representativeness, and random sampling. 7. Relate it to real school experiences, such as event planning, learner council surveys, or club promotions. Possible Answers for Reflection Questions Option 2: The Sampling Dilemma 1. Random sampling across strands and grades is more reliable. 2. Choosing similar people causes bias. 3. Randomness and diversity improve fairness and accuracy. Strategy 3: Inquiry-Based Group Work – Survey Fail Detective Instructions: 1. Provide learners with the survey scenarios
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	Group B: Randomly picked learners from every section Group C: Only surveyed learners who volunteered Instructions: 1. As a group of “survey detectives,” analyze each method. 2. Decide which is the best and which is the worst and explain why. 3. Present your findings using a poster or slide. Reflection Questions: 1. Which method gives the fairest results? 2. Which method might be biased? 3. How does random selection help ensure fairness in a survey or study?	2. Assign them the role of data detectives, encouraging critical thinking 3. Allow creative presentation formats (digital slides, skits, posters). 4. Guide reflection on sampling fairness, volunteer bias, and the importance of inclusion. 5. Guide reflection on sampling fairness, volunteer bias, and the importance of inclusion. 6. Connect the activity to real research practices or learner-led initiatives. Possible Answers for Reflection Questions Option 3: Survey Fair Detective 1. Random sampling from all sections is fairest. 2. Sampling only boys or volunteers is biased. 3. Random selection gives everyone an equal chance.
B. Instituting New Knowledge	B.1. Presenting Examples	
	Activity B.1: Exploring Key Concepts Instructions: Read each scenario and answer the questions that follow. Be prepared to share your thoughts with the class. Scenario 1: School Survey The learner council wants to know which school event to organize next semester. They decide to ask only 100 out of 1,000 learners. Questions: 1. What type of sampling technique could they use to choose fairly? 2. Why didn't they ask everyone? 3. What could make the results biased? Scenario 2: Research Project	Activity B.1: Exploring Key Concepts This activity introduces key sampling concepts through familiar and practical situations. Learners apply critical thinking to connect real-world cases with sampling methods, setting the stage for deeper conceptual understanding. Use any or all examples as discussion starters. To facilitate this activity, follow the instructions below. 1. Present the scenarios to the class using a printed worksheet, slide, or digital format. 2. Allow learners to answer individually or in small groups 3. Use guiding questions to draw attention to: • The definition and type of sampling used. • The practicality of sampling over full population studies. • How sampling affects bias and accuracy

	You are conducting a research project about learners' screen time. You collect data from learners in your own section. Questions: 1. Is this a good representation of the whole grade level? 2. What type of sampling method is being used in the scenario? 3. How could this sampling method be improved? Scenario 3: Barangay Health Survey The barangay health center wants to know the average weight of residents aged 18–60. They select every 5th person on the health registry. Questions: 1. What sampling method is this? 2. Is this considered random sampling? 3. Why might this method be efficient?	4. Lead a whole-class discussion to clarify the different sampling techniques and link them to key terms such as population, sample, statistic, parameter, random sampling, and bias. Answer Key: Scenario 1: School Survey 1. Simple random sampling or stratified sampling 2. It takes too much time and resources 3. Choosing learners from only one group Scenario 2: Research Project 1. No 2. Convenience sampling 3. Select learners randomly from different sections Scenario 3: Barangay Health Survey 1. Systematic sampling 2. Yes 3. It is quick, easy, and still representative
B.2. Discussing the Concept		
	Activity B.2: Deepening Conceptual Understanding Instructions: 1. Carefully read the lesson notes on sampling. 2. Take note of important terms. 3. Work through the step-by-step examples provided 4. Be ready to answer questions or share your thoughts during discussions. 5. Raise questions if their concepts are confusing. The Sampling Sampling is the process of selecting a subset (called a sample) from a larger group (called a population) to gather information and draw conclusions without examining the	Activity B.2: Deepening Conceptual Understanding This activity introduces the essential concepts of sampling aligned with the objectives. Prior to the discussion be reminded on the following: 1. Given the number of concepts to be covered, carefully plan your instructional approach to ensure learners can engage with the lesson without feeling overwhelmed. 2. Use visual aids like graphs and shaded areas to strengthen understanding. 3. Encourage learners to verbalize reasoning and explain steps to peers. 4. Integrate technology where possible.

	entire population. This activity introduces key sampling concepts such as random sampling, sample size, parameters vs. statistics, and the comparison between probability and non-probability sampling methods. Determining Appropriate Sample Size In sampling, the first thing that you must consider is the sample size. In any research, you usually cannot study the entire population because it may be too large, expensive, or time-consuming. Instead, researchers select a sample that represents the population. You must think: “How many people or items should you include in your study to obtain useful results? Having enough participants to observe trends or patterns should always be considered in an investigation to avoid wasting time or resources.” Factors Affecting Sample Size Even without statistical formulas, sample size depends on: <table border="0"> <tr> <td>Purpose of the study</td><td>:</td><td>Are you just exploring opinions, or do you want to compare groups?</td></tr> <tr> <td>Population size</td><td>:</td><td>How many people or items exist in total?</td></tr> <tr> <td>Resources available</td><td>:</td><td>Time, money, or workforce may limit how many participants you can include</td></tr> <tr> <td>Type of research</td><td>:</td><td>Surveys, experiments, or qualitative interviews each have different practical requirements.</td></tr> </table>	Purpose of the study	:	Are you just exploring opinions, or do you want to compare groups?	Population size	:	How many people or items exist in total?	Resources available	:	Time, money, or workforce may limit how many participants you can include	Type of research	:	Surveys, experiments, or qualitative interviews each have different practical requirements.	- Continuously relate abstract concepts to tangible, familiar examples. - The strategies provided are merely suggestions; you may apply the approach you believe best suits your learners during the discussion. - Ask check-in questions every concept and reflection question every end of the lesson. To facilitate: - Start by engaging learners with examples of surveys or polls they know. For example, in the Philippines, during presidential elections, polling firms like Pulse Asia or Social Weather Stations (SWS) conduct surveys by randomly selecting around 1,200 respondents from across the country. These respondents are interviewed about their preferred candidate. Because the sample is carefully chosen to reflect the diversity of the population, the results often give a fairly accurate prediction of the election outcome. - Define key terms: population, sample, parameter, statistic, random sampling. - Explain the importance of choosing the right sample size and sampling method. - Compare probability and nonprobability sampling with real-life examples. - Guide learners through the problems, encouraging group discussion. Problem 1: A school has 500 learners. You want to know their favorite snack but can only survey 50 learners. How will you select your sample to ensure it represents the entire school fairly? What sampling method would you use? Problem 2: Using the snack survey example, explain the difference between a parameter and a statistic.
Purpose of the study	:	Are you just exploring opinions, or do you want to compare groups?												
Population size	:	How many people or items exist in total?												
Resources available	:	Time, money, or workforce may limit how many participants you can include												
Type of research	:	Surveys, experiments, or qualitative interviews each have different practical requirements.												

Simple Ways to Decide Sample Size

A. Rule-of-Thumb Estimates

These rules-of-thumb are based on experience, simulation studies, and conventional practices in research, rather than a single published formula.

The purpose is to provide quick, practical estimates for researchers and educators, especially for teaching or initial planning, before using formal formula.

Type of Research	Suggested Sample Size
Survey (large population)	200–400 respondents for a general idea
Small group survey	At least 20–50 respondents
Experimental research (2 groups)	15–30 participants per group
Qualitative research (interviews)	10–20 participants, or until no new information emerges (saturation)

B. Using a Simplified Formula

If the population is known and small, you can use a simple “proportion-based” formula:

$$\text{Sample size} = \frac{\text{Population}}{2 \text{ or more}}$$

Example: A class has 60 learners. A teacher wants to survey about class preferences. A simple choice is 30 learners (half of the class). If more precision is needed, you can increase the sample to 2/3 of the population.

C. Using Technology

You can use online sample size calculators or spreadsheet tools:

Which one applies to the whole school and which one to your sample?

6. Highlight why random sampling improves fairness and accuracy.

This lesson sets the foundation for calculating sample sizes and distinguishing between sampling methods in subsequent lessons.

The teacher can also use the video from DepEd TV: Statistics and Probability – Sampling Techniques (<https://www.youtube.com/watch?v=5PsF5MsrCOo>)

- | | | |
|--|---|--|
| | <p>Online calculators Just enter population size and desired margin of error</p> <ul style="list-style-type: none"> • Raosoft Sample Size Calculator • SurveyMonkey Sample Size Calculator <p>Spreadsheet method Create a column of participants, then randomly select a number based on available resources or rules of thumb.</p> | |
|--|---|--|

The Random Sampling

Random sampling means that every individual in the population has an equal chance of being selected. This method is important because it:

- Reduces bias: No one is unfairly favored or left out.
- Increases representativeness: The sample better reflects the diversity of the whole population.
- Improves reliability of results: Conclusions drawn from a random sample are more likely to be valid for the whole population.

Example: If you have 500 learners and want to select 50 randomly, you might use a computer to generate 50 random numbers representing learner IDs.

Types of Sampling

Probability Sampling

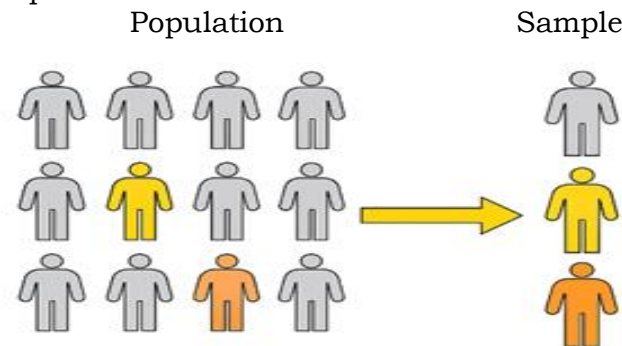
Every individual has a known chance of being chosen. Probability sampling is generally preferred because it leads to more accurate and generalizable results, while nonprobability sampling can lead to bias.

Examples include:

- Simple Random Sampling:**

Everyone is equally likely to be picked (e.g. randomly drawing names or using a random number generator).

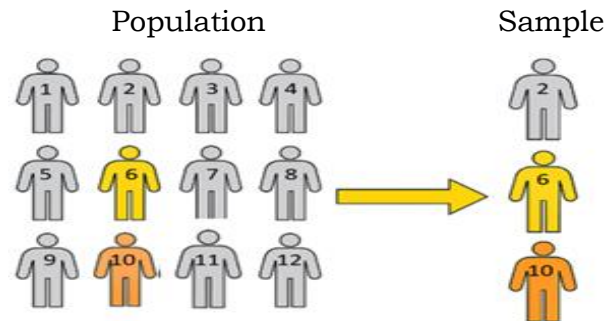
For example, in a population of 200 Grade 11 learners, the researcher needs a sample of 40 learners. Using simple random sampling, every learner in the population has an equal chance of being chosen. The researcher assigns numbers from 1 to 200 to all learners and then uses a random method, such as drawing lots or using a random number generator, to select 40 unique numbers. The learners correspond to those numbers form the sample.



- b. Systematic Sampling:
Selecting every k th individual from a list (e.g., every 4th learner). The starting point is chosen at random.

For example, in 200 Grade 12 learners, you will choose 50 learners as members of your sample. To select which learners will be part of the sample, divide the population size 200 by the sample size 50, which results to 4. That means every 4th learner among the 200 Grade 12 learners will be included in the sample. You may also start on any randomly selected number from 1 to 4; for instance, if the starting number is 2, then the sample will include

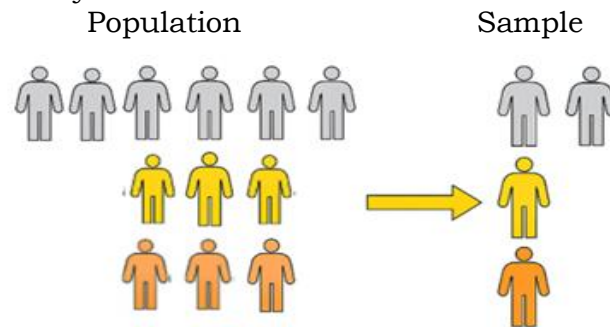
learners numbered 2, 6, 10, and so on, until the desired 50 learners are chosen. (Note: There are other patterns.)



- c. **Stratified Sampling:**
The population is divided into subgroups or strata (e.g., by grade level or strand). Samples are randomly selected from each stratum to ensure all groups are represented.

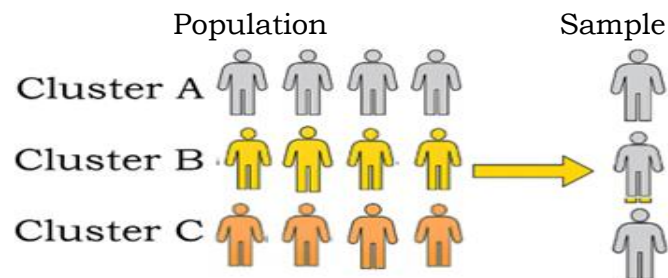
For example, in a population of 300 Grade 11 learners composed of three strands: 150 STEM learners, 90 HUMSS learners, and 60 GAS learners, the researcher needs a sample of 60 learners. To make sure each strand is fairly represented, the researcher uses stratified sampling. First, the proportion of each group is determined: STEM is $150/300 = 50\%$, HUMSS is $90/300 = 30\%$, and GAS is $60/300 = 20\%$. Next, these same proportions are applied to the sample size of 60. Thus, the researcher will select 30 STEM learners (50% of 60), 18 HUMSS learners (30% of 60), and 12 GAS learners (20% of 60).

Finally, the learners are chosen until the required number of learners from each strand is then selected randomly.



- d. Cluster Sampling: The population is divided into clusters, usually based on location or existing groups (e.g., classes or barangays). A few clusters are randomly selected, and all individuals within those clusters are surveyed.

For example, in a population of 150 Grade 11 learners, the researcher needs a sample of 50 learners. The 150 learners are already divided into 3 sections with 50 learners each. Using cluster sampling, the researcher randomly selects 1 section (cluster) out of the 3. All 50 learners in the chosen section automatically become the sample.



Cluster A is randomly chosen as sample

Nonprobability Sampling

Selection is based on convenience or judgment, not chance. Examples include:

- a. Convenience Sampling
Choosing whoever is easiest to access (e.g., the first 50 learners who enter the cafeteria)
- b. Purposive (Judgmental) Sampling
Choosing individuals who possess specific characteristics relevant to the study.

Example: Interviewing only science teachers when studying teaching strategies in science.

- c. Quota Sampling
The population is divided into groups, and a predetermined number of participants is selected from each group, though the selection is non-random.

Example: Surveying 20 male and 20 female learners to reflect gender balance.

- d. Snowball Sampling
This technique relies on existing participants to recruit additional participants, creating a referral chain.

Example: A learner respondent recommends friends, who in turn recommend others, to join the study.

- e. Voluntary Response Sampling
In this type, individuals volunteer or decide by themselves to participate in the research.

Example: Learners who willingly respond to an online survey posted by the school.

	Parameter vs. Statistic These two terms describe similar measurements but at different levels: Parameter: A value that describes a characteristic of the entire population (e.g., the average snack preference of all 500 learners). This is usually unknown because it's hard to collect data from everyone. Statistic: A value that describes the same characteristic but based on the sample (e.g., the average snack preference of the 50 learners surveyed). Statistics are used to estimate parameters. The goal of sampling is to use statistics to make reasonable guesses about parameters.	
B.3. Developing Mastery		
	Activity B.3: Practicing Learned Skills Instructions: Read each activity carefully and perform the task as directed. Option 1: Classroom Simulation Part I. Fill-in-the-Blank Sampling! Instructions: Complete each sentence. 1. A _____ is the entire group being studied 2. A _____ is a smaller part of the population 4. A _____ is a characteristic of a sample. (Statistic) 5. _____ means everyone has an equal chance of being chosen. Part II. Show Us What You Got! Instructions: 1. Each group will act out one sampling method using classmates or name cards.	Activity B.3: Practicing Learned Skills This activity promotes active learning by engaging learners in realistic, contextual problems. You may select any of the provided activities to enhance learners' understanding of the lesson on sampling. Option 1: Classroom Simulation Instructions: Part I. 1. Organize learners into small groups. 2. Distribute whiteboards or index cards and markers. 3. Read each fill-in-the-blank question aloud, one at a time. 4. Say: "Work together in your group. When I say 'Show,' raise your answers at the same time." 5. After each response, confirm the correct answer and provide a short explanation. Part II.

	2. Decide if it is probability or nonprobability sampling. 3. Explain why sample size and method matter. Does your method give accurate results? Is it quick and easy, or fair and representative? Part III. Sample It Right! Instructions: 1. Each group will be provided with scenario cards describing different sampling situations a. A teacher wants to know the preferred mobile phone brand of all Grade 11 learners in your school. Every learner should have an equal chance of being selected. b. The librarian wants to survey learners about their reading habits. She lines up all the learners who enter the library in one week and selects every 5th learner to answer the survey. c. The learner council wants to survey 100 learners about their study habits. Since the school has three strands (STEM, HUMSS, GAS), they want to make sure each strand is fairly represented in the sample. d. A barangay health worker wants to survey families about waste management. Instead of selecting households from all over the barangay, she randomly chooses two whole streets and surveys every household in them. e. A learner researcher wants quick feedback about the cafeteria food. He surveys only the learners sitting at the nearest tables during lunchtime. f. A guidance counselor wants to understand the experiences of learners with anxiety.	1. Assign each group one sampling method: Simple Random, Systematic, Stratified, Cluster, Convenience, Snowball, Purposive, or Quota 2. Provide name cards and a container for demonstrations. 3. Instruct groups to demonstrate their assigned method. 4. After each demonstration, ask the group: • Is this probability or non-probability sampling? • Why do the sampling method and sample size matter? • Is the method accurate and fair, or mainly quick and easy? 5. Summarize the strengths and weaknesses after each presentation Part III. 1. Distribute scenario cards, task sheets, and chart paper. 2. Say: “Read your scenario carefully. As a group, decide the most appropriate sampling method and possible sample size. Answer the Reflection Question and write your plan on the chart paper.” 3. Allow groups to present their sampling plan. 4. Give feedback and connect responses to key principles: representativeness, feasibility, bias, and purpose of the investigation. Answer Key Part I 1. Population 2. Sample 3. Parameter 4. Statistic 5. Random Sampling
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- She interviews one learner, who then refers her to another learner with the same concern, and the process continues.
- g. A science teacher wants to study how top-performing learners prepare for exams. She chooses only the honor learners from each section to participate in the survey.
 - h. A researcher wants to survey 60 learners about their social media habits. To ensure balance, she decides to collect responses from exactly 20 STEM, 20 HUMSS, and 20 GAS learners, regardless of how many are available.
- The group will identify the population, choose an appropriate sampling method, and decide on sample size.
 - Your group will present the sampling strategies and justify your choices.

Reflection Questions:

- What is the target population?
- Which sampling method fits this scenario best?
- How will you ensure the sample is representative?
- How large should the sample be for reliable results?
- What challenges might arise in your sampling method?

Part II

- Simple Random: Fair, accurate, but time consuming.
- Systematic: Easy, may skip patterns.
- Stratified: Covers all subgroups.
- Cluster: Saves time, less representative.
- Convenience: Quick, but biased.
- Snowball: Finds hidden groups, not general.
- Purposive: Targeted, but subjective.
- Quota: Balances subgroups, no randomness

Part III

	Population	Method	Sample	Challenge
a.	All Grade 11 learners	Simple Random	50–100 learners, equal chance	Time-consuming, needs full list
b.	Library users	Systematic	Every 5th learner (30–50)	May miss hidden patterns
c.	STEM, HUMSS, GAS learners	Stratified	100 learners, proportional per strand	Must know exact subgroup sizes
d.	All barangay households	Cluster	All households in 2 chosen streets	Clusters may not represent whole
e.	Cafeteria users	Convenience	20–30 nearby learners	Biased, not representative
f.	Learners with anxiety	Snowball	Grows by referrals	Limited, non-representative
g.	Honor learners	Purposive	All top achievers per section	Subjective, excludes others
h.	STEM, HUMSS, GAS learners	Quota	20 per strand (60 total)	Not random, may misrepresent

Option 2: **Sampling Challenge**

Part I. Matching Race

Instructions:

1. Work with a partner.
2. You will receive two sets of cards: terms and definitions.

Terms	Definitions
○ Parameter	○ The entire group being studied
○ Population	○ A smaller part of the population
○ Random Sampling	○ A characteristic of the population
○ Sample	○ A characteristic of the sample
○ Statistic	○ Everyone has an equal chance of being chosen

3. Your task is to match each term with its correct definition
4. You have 2 minutes—work quickly, it's a race!
5. The class will check answers together. Fastest correct group earns points or praise.

Part II. **Case Challenge**

Instructions:

1. Read each case scenario carefully.
2. Identify which sampling method is being used.
3. Work in pairs or small groups to discuss your answers.

Case Scenarios:

1. A teacher wants to know the learning style of all A learner quickly surveys classmates in the canteen.
2. Grade 11 learners.
A clinic wants to survey learners with asthma.
3. A researcher wants fair representation from all Track (Academic, TechPro).

Option 2: **Sampling Challenge**

Instructions:

Part I.

1. Provide each pair/group with two sets of cards: Terms (Population, Sample, Parameter, Statistic, Random Sampling) and Definitions.
2. Say: "Match each term with its correct definition as quickly as you can. You have 2 minutes—it's a race!"
3. After time is up, review the matches together. The fastest correct group earns points or praise

Part II.

1. Distribute the case scenario cards to each group.
2. Say: "Read each case carefully. Discuss with your group and identify which sampling method is being used."
3. After discussion, groups share their answers, while the teacher confirms and explains briefly.

Part III.

1. Say: "Look back at the five case scenarios. Decide which are probability sampling and which are nonprobability sampling. Work with your group and be ready to explain your choices."
2. Display the reflection questions:
 - Which sampling method do you think gives the most reliable results, and why?
 - Which method would you personally use if asked to survey all Grade 11 learners in your school?
3. Facilitate discussion, clarifying key differences between probability and nonprobability methods

4. A barangay officer surveys whole blocks of houses.

Part III. **Sampling Showdown**

Instructions:

1. Look back at the five case scenarios from Part II.
2. Decide which ones are probability sampling and which are nonprobability sampling.
Share your answers with your partner.
3. Be ready to explain your choices during the class discussion.
4. Reflect:
 - Which sampling method do you think gives the most reliable results, and why?
 - Which method would you personally use if you were asked to survey all Grade 11 learners in your school?

Option 3: **Sampling Face-Off!**

Part I. **Hot Seat Game**

Instructions:

1. One chair in front is the “hot seat.”
2. Volunteers (or chosen learners) will sit in the hot seat one at a time.
3. The teacher will ask a question. If you are in the hot seat, you must answer within 10 seconds.
4. If correct, you earn a point for your group. If not, the chance passes to another learner.

Hot Seat Questions

1. What is a parameter?
2. What is a statistic?
3. What is random sampling?
4. What is the difference between population and sample?
5. Why is random sampling important?

Part II. **Face-Off**

Answer Key

Part I

1. Population → The entire group being studied
2. Sample → A smaller part of the population
3. Parameter → A characteristic of the population
4. Statistic → A characteristic of the sample
5. Random Sampling → Everyone has an equal chance of being chosen

Part II

1. Learning styles → Simple Random (probability)
2. Learners with asthma → Purposive (nonprobability)
3. Track representation → Stratified (probability)
4. Whole blocks of houses → Cluster (probability)
5. Canteen classmates → Convenience (nonprobability)

Part III

Probability: Simple Random, Stratified, Cluster

Nonprobability: Purposive, Convenience

Option 3: Sampling Face-Off!

Instructions:

Part I.

1. Place one chair in front of the class as the “Hot Seat.”
2. Call for volunteers (or select learners) to take turns sitting in the hot seat.
3. Say: “I will ask you a question. You have 10 seconds to answer. If you get it right, your group earns 1 point. If not, another learner gets the chance.”
4. Continue until all questions are answered.

Part II.

1. Divide the class into two teams:
 - Team A → Defends Probability Sampling (accurate, representative).

	Instructions: - The class will be divided into two teams: - Team A - Argues for Probability Sampling - Team B - Argues for Nonprobability Sampling - Each team will have 2 minutes to prepare arguments. - Debate Rules: - Opening Arguments: Each team speaks for 3 minutes. - Rebuttals: Each team has 2 minutes to respond to the other side. - Q&A: Teams may ask each other questions for 2 minutes. - Be respectful, listen carefully, and support your answers with reasons or examples. Part III. Reflect Instructions: - Take 5 minutes to write a short reflection in your notebook. - Answer these questions: - Why is choosing the right sample size and method important? - Which sampling method would you choose for school-based research? Why? - Be ready to share your answer if the teacher calls on you.	- Team B → Defends Nonprobability Sampling (faster, practical). - Give each team 10 minutes to prepare their arguments. - State debate rules: - Opening Arguments: Each team speaks for 3 minutes. - Rebuttals: Each team responds for 2 minutes. - Q&A: Teams may ask each other questions for 5 minutes. - Remind learners: “Stay respectful, listen actively, and support your points with reasons or examples.” Part III. - On learners notebooks, they will write short responses to: - Why is choosing the right sample size and method important? - Which sampling method would you choose for school-based research? Why? - Invite a few volunteers to share their reflections in the class discussion. Answer Key Part I - Statistic → A characteristic of the sample. - Random Sampling → Everyone has an equal chance of being chosen. - Importance of Random Sampling → Ensures fairness and reduces bias. - Population vs. Sample → Population is the entire group; sample is a smaller part of it. Part II Probability Sampling Strengths: Representative, fair, reliable results.
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		Probability Sampling Weaknesses: Time-consuming, requires resources. Nonprobability Sampling Strengths: Quick, practical, useful in limited situations. Nonprobability Sampling Weaknesses: Biased, less generalizable Part III Right sample size and method ensure accuracy, fairness, and reliability. Many learners may prefer Stratified or Simple Random Sampling for school-based research because they are fair and representative.
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Making Real-World Connections Instructions: Read carefully the scenario and do the tasks as indicated. Scenario: Your school is preparing for a Cultural Celebration Day. The goal is to make sure all tracks feel included the Academic and TechPro. The organizers need to collect learner opinions using a sampling plan that is fair and representative of the whole school. Case 1. Talent Show Line-up The school wants to know what kind of performances learners prefer for the Talent Show. If the survey only asks learners from the Academic track, the results may not reflect the interests of TechPro. Create a sampling plan that represents all tracks fairly. Case 2. Limited Time The event is next week, and the organizers don't have time to survey all 600 learners. They decide to pick sections to answer the survey. If they only choose	Activity C.1: Making Real-World Connections This activity encourages learners to think critically about how they applied sampling concepts in the earlier scenario-based task. Through guided reflection, students examine the reasoning behind their choices, consider fairness and accuracy, and appreciate the practical importance of sampling in real-life situations such as school events, surveys, and decision-making. By answering the reflection prompts, learners connect classroom knowledge to real-world applications, deepen their understanding of representative sampling, and recognize how proper planning can save time and resources while ensuring reliable results. Instructions: - Set the stage - Read aloud the activity prompt and scenario overview: the school will hold a Cultural Celebration Day and needs a fair sampling plan.

	sections from one track, the results won't be fair. Decide which sections to include so that the survey results represent all tracks. Case 3: Food Fair During the celebration, there will be a Food Fair. Learners from different tracks may have different food preferences. If the survey is done only in one track, the results will not capture everyone's opinion. Make a sampling plan that ensures all track's choices are included. Task: Choose only one the cases and design a sampling plan to collect information from a portion of the learner population. This plan should be fair, unbiased, and representative of the school community. Work together to figure out: • Who to ask? • How to pick them • How many to ask • How to make sure your survey is fair and accurate Answer the questions below. Work quickly but think carefully. Be ready to share your ideas with the class. Identify the Population Who is your target population? Who should be included in your survey? Explain why you chose this group. Choose a Sampling Method What method will you use to choose your sample? Why did you choose this method? Estimate the Sample Size Suppose your school has 600 learners. To get information, you want to survey enough people. How many learners will you survey? Explain your answer.	• Explain the group task: choose one case and design a sampling plan answering the provided questions 2. Group work • Give each group a case card and task sheet (or let them choose one case). • Tell groups to appoint a recorder and a presenter. • Circulate while groups work: ask probing questions (see "Teacher prompts" below) and check that groups identify population, method, sample size, and fairness issues. 3. Presentations • Each group presents the sampling plan (2 minutes per group). Ask the presenter to show their chart paper and answer the Reflection prompts. • After each presentation, give short feedback (1–2 minutes): confirm correct points, highlight missed issues, and connect to sampling concepts 4. Wrap up (2 minutes) • Reinforce main takeaways: population vs sample, why sampling method and size matter, and simple ways to reduce bias. • Optionally, ask students to reflect in one sentence: "One thing I learned about sampling is..." 5. Reflection Questions • How did your group decide on the sampling method for your scenario? • What made your sample fair and representative of the whole school? • Why is it useful to use sampling instead of asking everyone in real-life situations? • How could your sampling plan affect the accuracy of the results?
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	Consider Fairness Suppose your school has 600 learners. To get information, you want to survey enough people. - What problems or bias might happen in your sampling plan? - How can you reduce or avoid these problems?	Can you think of other real-life situations where sampling would save time and still give useful information?			
	Twist The school has: 300 boys, 300 girls Academic = 450 TechPro = 150 How would this information affect your sampling method and sample size?	Criteria	Excellent (3 pts)	Good (2 pts)	Needs Improvement (1)
		Population Identified	Population Identified	Clear and complete	Mostly clear
		Sampling Method Chosen	Sampling Method Chosen	Appropriate and well explained	Chosen with basic explanation
		Sample Size Estimated	Sample Size Estimated	Realistic and justified	Somewhat reasonable
		Bias Considered	Bias Considered	Bias identified and solution offered	Some bias considered
		Clarity of Plan	Clarity of Plan	Very clear and easy to understand	Mostly clear
C.2. Making Generalization					
Activity C.2: Wrapping up the Lesson Option 1: Reflection Instructions: 1. Write short notes in your notebook using the Reflection Question below: - What is random sampling? Name at least two types (e.g., simple random, stratified). - How do you calculate or decide the appropriate sample size for a study? - What is the difference between a parameter and a statistic? - How are probability and nonprobability sampling techniques different in terms of fairness and use?		Activity C.2: Wrapping up the Lesson To wrap-up the lesson, you may apply one of the following strategies. These activities reinforce key concepts in sampling, ensuring learners can connect theory to application and demonstrate understanding in either written or visual form. To wrap-up the lesson you may facilitate any of the suggested strategies. Strategy: 1 Reflection Writing Instructions: 1. Ask learners to take out their notebooks and pens.			

2. After answering, write a short summary paragraph (4–5 sentences) explaining how a good sample leads to accurate generalization about a population

Option 2: Concept Map

Instructions:

1. Draw a central circle labeled “Sampling and Generalization.”
2. Create four branches with these labels:
 - Random Sampling & Types → write examples (simple, stratified, cluster, systematic).
 - Sample Size → show how many learners out of 600 you might survey.
 - Parameter vs. Statistic → give a short example (parameter = average height of all learners, statistic = average height of 50 sampled learners).
 - Probability vs. Nonprobability Sampling → add 1–2 keywords for each (probability = fair, unbiased; nonprobability = quick, practical).
3. Connect your branches back to the main circle to show how these ideas help in making generalizations from sample to population

2. Display or write the Reflection Questions on the board/projector.
3. Instruct: “Answer the Reflection Question briefly in your notebooks.” Allow 5–7 minutes.
4. Write a 4–5 sentence summary explaining how a good sample leads to accurate generalization about a population.
5. Invite one or two volunteers to share their summaries aloud. Conclude by emphasizing the importance of representative sampling

Strategy 2: Concept Mapping

Instructions:

1. Distribute blank paper/chart paper and markers (or ask learners to use notebooks if supplies are limited).
2. Instruct: “Draw a central circle labeled ‘Sampling and Generalization.’”
3. Guide learners in building their concept map by creating four branches:
 - Random Sampling & Types → simple, stratified, cluster, systematic
 - Sample Size → example: 100 out of 600 learners
 - Parameter vs. Statistic → parameter = population average; statistic = sample average
 - Probability vs. Nonprobability → probability = fair/unbiased; nonprobability = quick/practical
4. Remind learners to connect all branches back to the main circle.
5. Allow 7–10 minutes for completion, then ask a few learners to share their maps. Wrap up by highlighting how these concepts work together in making accurate generalizations.

C.3. Evaluating Learning

Activity C.3: Assessing Learning Outcomes

Instructions:

Read each direction carefully and perform the tasks as indicated.

Part I. Multiple Choices

Write the letter of the best answer.

Scenario for items 1 and 2:

The Learner Council wants to know learners' favorite lunch meal to improve the canteen menu. There are 900 learners in the school.

1. To make the results fair, which method should the Learner Council use?
 - a. Ask only the council officers
 - b. Ask friends during lunch
 - c. Use random sampling so every learner has an equal chance
 - d. Post a survey only on social media
2. If they decide to survey 10% of the learners, how many should they include in the sample?
 - a. 45
 - b. 90
 - c. 100
 - d. 120

Scenario 2 for items 3 and 4.

The principal wants data about all Grade 11 learners' favorite sports. A teacher collects data from one section only and claims it represents all Grade 11 learners.

3. The teacher's result is an example of:
 - a. Parameter
 - b. Statistic
 - c. Population
 - d. Sample size
4. Why might the teacher's result not represent all Grade 11 learners?

Activity C.3: Assessing Learning Outcomes

This activity evaluates learners' knowledge and application of sampling concepts, including random sampling, sample size, parameters, statistics, and the distinction between probability and nonprobability sampling.

Instructions:

1. Prepare Materials
 - Printed quiz sheets or displayed questions (board/projector)
 - Answer sheets or learner notebooks
 - Pens or pencils
2. Instruct learners:
 - "Read each the instructions carefully."
3. Allow learners to accomplish the tasks.
4. Collect responses for assessment or invite volunteers to share their answers orally.
5. Gather feedback from the learners by asking the Reflective Questions:
 - Which question did you find the most challenging, and why?
 - How did the lesson on sampling help you understand the scenario better?
 - Can you think of a real-life situation where you might use what you learned in this activity?

Answer Key

Part I.

1. c 2. b 3. b 4. b 5. a

Part II.

1. Random sampling is important because it ensures fairness, minimizes bias, and allows all members of the population an equal chance of selection.

	- Because it is too large - Because only one section was included - Because it uses random sampling - Because it surveyed the whole population Scenario for item 5. A barangay health worker wants to survey high school learners about their sleeping habits. She considers two options: Option A (Probability): Randomly selects 100 learners from the master list. Option B (Nonprobability): Asks 100 learners near the covered court after class. - Which option is more likely to represent the entire learner population? - Option A, because every learner had an equal chance of being chosen - Option A, because it only uses volunteers - Option B, because it is easier and faster - Option B, because it includes only active learners Part II. Open-Ended (Application & Reasoning) Answer briefly and clearly in 2–3 sentences. - In the school canteen survey, why is random sampling important for fairness? - In the sports festival example, explain the difference between a parameter and a statistic. Give one example of each. - In the community health project, when might the health worker prefer nonprobability sampling (Option B) even if it is less accurate? - Compare probability and nonprobability sampling. Which one would you use for research in your school? Explain why.	- A parameter describes a population (e.g., the favorite sport of all Grade 11 learners), while a statistic describes a sample (e.g., the favorite sport of one section). - A health worker may prefer nonprobability sampling if time, resources, or access to a full population list are limited. - Probability sampling is representative and accurate, while nonprobability sampling is quicker but less reliable. For school-based research, probability sampling is recommended to ensure valid and reliable results
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C.4. Additional Activities

Activity C.4: Extending and Reinforcing Learning

Instructions:

Read each direction carefully and perform the tasks as indicated.

Remediation Activity

Part A. Sampling Match-Up

Match each situation with the correct sampling method from the box. Write only the letter of your answer in the blank.

- A. Convenience Sampling
 - B. Stratified Sampling
 - C. Cluster Sampling
 - D. Simple Random Sampling
 - E. Systematic Sampling

1. A researcher surveys every 5th learner on the enrollment list. ____
2. A health worker randomly selects 50 learners from a master list. ____
3. A teacher chooses one section and surveys all learners in it. ____
4. A learner asks only his close friends for a quick survey. ____
5. A researcher divides learners by strand (STEM, HUMSS, GAS) and selects proportionately. ____

Part B – Real-life Scenario

Read each situation carefully, solve, and explain your answer.

1. A school has 1,200 learners. If the researcher wants to survey 10%, how many learners should be included in the sample?

Activity C.4: Extending and Reinforcing Learning

This activity reinforces learners' understanding of sampling methods, sample size computation, and practical application through real-life scenarios. It also develops higher order thinking by asking learners to design their own sampling plan.

Instructions:

1. Prepare Materials
 - Printed activity sheets.
 - Answer sheets
2. Based from the results of the evaluation, assigned the appropriate activity for the learners and distribute the activity sheet.
3. Allow learners to accomplish the tasks at home.
4. Check learners answer to track their progress.

Answer Key:

Remediation

Part I.

1. E 2. D 3. C 4. A 5. B

Part II.

1. 120
2. 40
3. Cluster Sampling – groups (blocks) chosen instead of individuals

Enrichment Activity

1. Target population: all learners (or specified group, if justified).
2. Method: stratified or simple random sampling (ensures fairness).
3. Sample size: 10% of 900 = 90 learners.

	2. In a population of 800 learners, 20% prefer online learning. If a sample of 200 learners is taken, how many are expected to prefer online learning? 3. A barangay has 10 blocks of houses. A researcher randomly chooses 3 blocks and surveys all households in them. • What type of sampling is this? • Why is it classified under this method? Enrichment Activity Read the scenario and perform the tasks. Scenario: Your school is preparing a Wellness Program to support learners’ mental and physical health. The goal is to make sure the survey represents the whole learner body fairly. Answer each question. 1. Who is your target population? 2. Which sampling method will you use? Why? 3. Suppose your school has 900 learners. How many will you include in your sample? 4. What possible biases or problems might occur in your plan? 5. How will you avoid these problems to make sure your results are fair and useful? Write a short report (5–7 sentences) summarizing your plan, including: • Introduction (purpose of the survey) • Method (technique, sample size, justification) • Reflection (importance of fairness in sampling)	4. Possible biases: convenience, nonresponse, limited coverage. 5. Solutions: random selection, reminders, multiple data collection methods.
VI. SUMMATIVE ASSESSMENTS	Please refer to the attachments for the summative assessment and performance task. Answer Key — Part I: Multiple Choice 1. D 6. D 11 D 16. A 2, A 7. B 12. C 17. A	

	3. D 8. B 13. C 18. B 4. C 9. A 14. A 19. A 5. D 10. D 15. B 20. A Part II: Real-life Situations A. SHS Intramurals Planning 1. Sampling method: Stratified sampling (selecting 25 learners from each track). 2. Why appropriate: Ensures each SHS track is represented, allowing comparisons across tracks and reducing subgroup sampling error. 3. If only Academic learners surveyed — issue: Results would be biased and not generalizable to other tracks. 4. One strength / one limitation: - o Strength: Provides proportional representation of subgroups (improves representativeness). - o Limitation: Requires accurate lists of strata and more coordination/logistics. 5. 10% of 1,200: 120 learners. B. Social Media Study 6. Sampling method: Convenience sampling. 7. Representative? No — the sample is biased (limited to the researcher’s class) and does not reflect the full SHS population. 8. Parameter vs. Statistic: - o Parameter: A value describing the entire population (unknown), e.g., average hours all SHS learners spend on social media per day. - o Statistic: A value computed from the sample, e.g., average hours from the 50 surveyed learners. 9. Technique to improve reliability: Use simple random sampling or stratified sampling (and/or increase sample size) to obtain a more representative sample. 10. Why random sampling is important: It reduces selection bias, gives each unit an equal chance of selection, and supports valid generalization and inference to the population
VII. REFLECTION	Reflection is most effective when done regularly. Make it a habit to reflect after each lesson or weekly to track your progress and growth over time. The reflection questions are designed to help you thoughtfully evaluate your teaching, identify strengths, recognize challenges, and plan improvements to enhance student learning. The questions provided below may help in reflecting to your teaching performance. 1. What aspects of your teaching went particularly well? • Which part of the lesson engaged students the most? • What teaching strategies or methods contributed to successful learning? • How did students respond to your instruction (e.g., participation, performance, behavior)? • What evidence shows that learning objectives were achieved?

- What personal teaching strengths were evident in this session?

2. What challenges did you encounter, and how did you address them?

- What difficulties arose during lesson delivery or classroom management?
- Were there moments when students struggled to understand a concept? Why?
- How did you adjust your strategies or pacing to respond to these challenges?
- What support or resources would have helped you handle the situation more effectively?
- What will you do differently next time to overcome similar obstacles?

3 What will you do differently next time to improve student learning?

- Which teaching strategies or approaches would you modify or try in the future?
- How will you better address diverse learning needs in your classroom?
- Are there new resources, activities, or technologies you plan to incorporate?
- What steps can you take to enhance student engagement or participation?
- How will you assess whether these changes improve learning outcomes?

Unit 9 - SUMMATIVE TEST-

Part I. Multiple Choices

Directions: Read each item carefully. Select the letter of the correct answer and write it on your answer sheet. Each item is worth 1 point. Use a z-table or calculator if needed.

1. Which of the following is a discrete variable?
 - A. Time taken to run a race
 - B. Height of learners in a class
 - C. Temperature in a city on a given day
 - D. Number of cars passing a checkpoint in an hour
2. A continuous variable is best described as:
 - A. A variable that can take any value within an interval
 - B. A variable that takes only whole number values
 - C. A variable that has only two possible outcomes
 - D. A variable measured by counting categories
3. The expected value of a discrete random variable X is:
 - A. The most frequently occurring value of X
 - B. The square root of the sum of the squared values of X
 - C. The middle value when X is ordered from least to greatest
 - D. The weighted average of all possible values of X, weighted by their probabilities
4. What is the variance of a discrete random variable?
 - A. the range of possible outcomes
 - B. the average value of the distribution
 - C. a measure of how spread out the values are
 - D. the difference between the highest and lowest values
5. How is the variance of a discrete random variable calculated?
 - A. by taking the square root of the mean
 - B. by finding the mode of the probability distribution
 - C. by multiplying each value by its probability and summing the results
 - D. by assuming the square roots of the differences between each value and the mean
6. What does the standard deviation of a probability distribution measure?
 - A. the average value of the distribution
 - B. the likelihood of each outcome occurring
 - C. the difference between the highest and the lowest values
 - D. the spread of the dispersion of the values around the mean
7. The standard deviation of a discrete random variable X is:
 - A. The variance squared
 - B. The expected value of X
 - C. The square root of the variance
 - D. The reciprocal of the variance
8. Suppose a discrete random variable X takes values 1, 2, and 3 with probabilities 0.2, 0.5, and 0.3 respectively. What is the expected value of X?
 - A. 1.0
 - B. 2.1
 - C. 2.5
 - D. 3.
9. Variance measures:
 - A. the spread or dispersion of the random variable values around the mean
 - B. the central tendency of a dataset
 - C. the highest value in the dataset
 - D. the average of the data values

10. In practical analysis, when might a continuous random variable be discretized?
 - A. when the distribution is uniform
 - B. when dealing with small sample sizes
 - C. when it is easier to measure in continuous form
 - D. when the data needs to be categorized for analysis
11. Which of the following is NOT a property of the normal distribution?
 - A. It is bell-shaped.
 - B. It is symmetric about the mean.
 - C. The mean, median, and mode are equal.
 - D. About 50% of the values lie within $\pm 1\sigma$ of the mean.
12. In a normal curve, approximately what percentage of data lies within two standard deviations of the mean?
 - A. 50%
 - B. 68%
 - C. 95%
 - D. 99.7%
13. A test has $\mu = 60$ and $\sigma = 10$. If a learner scores 80, what is the z-score?
 - A. 1.0
 - B. 1.5
 - C. 2.0
 - D. 2.5
14. If $z = -1.2$, $\mu = 50$, and $\sigma = 5$, what is the raw score (X)?
 - A. 44
 - B. 46
 - C. 54
 - D. 56
15. SAT scores are normally distributed with $\mu = 500$ and $\sigma = 100$. What is the probability that a learner scores below 600?
 - A. 0.6915
 - B. 0.8413
 - C. 0.9332
 - D. 0.9772
16. For a standard normal variable z , what is $P(z > 1.96)$?
 - A. 0.0250
 - B. 0.0500
 - C. 0.9750
 - D. 0.9900
17. The lifespans of lightbulbs are normally distributed with a mean of 1,000 hours and a standard deviation of 120 hours. What percentage of bulbs last longer than 1,200 hours?
 - A. 4.78%
 - B. 6.68%
 - C. 9.12%
 - D. 10.56%
18. Heights of learners are normally distributed with $\mu = 170$ cm and $\sigma = 6$ cm. What height corresponds to the top 2.5% of learners?
 - A. 180.8 cm
 - B. 181.8 cm
 - C. 182.8 cm
 - D. 183.8 cm
19. In a company, the weights of rice bags are normally distributed with a mean of 50 kg and a standard deviation of 2 kg. What percentage of bags weigh less than 47 kg?
 - A. 6.68%
 - B. 8.08%
 - C. 9.12%
 - D. 10.23%
20. Exam scores are normally distributed with a mean of 75 and a standard deviation of 8. What score corresponds to the bottom 10% of learners?
 - A. 64.7
 - B. 65.7
 - C. 66.7
 - D. 67.7

Part II: Real-life Situation

Directions: Read each situation carefully and answer the questions that follow. Write your answers clearly and concisely.

A. SHS Intramurals Planning

The school is preparing for the Senior High School intramurals. The Sports Committee wants to know what type of sports learners prefer to include in the program. Since they cannot ask every learner, they randomly select 25 learners from each SHS track:

Academic and TechPro

1. What sampling method is used in this situation?
2. Why is this sampling method appropriate for the purpose of the study?

3. If the committee surveyed only Academic track learners, what possible issue may arise with the results?
4. Identify one strength and one limitation of the sampling method used.
5. If the total SHS population is 1,200 learners, how many learners should be selected to represent 10% of the population?

B. Social Media Study

A learner researcher wants to find out how many hours SHS learners spend on social media per day. For convenience, the researcher surveys only the 50 learners in their own class.

6. What sampling method is used in this case?
7. Is this sample representative of the SHS population? Why or why not?
8. In this context, explain the difference between a parameter and a statistic.
9. What sampling technique that would improve the reliability of the findings.
10. Why is random sampling important when conducting survey

Unit 9 - Performance Task

Data-Driven Decisions: A Mini Statistical Investigation

Goal

You will design and carry out a mini statistical investigation about how many hours senior high school learners spend on social media each day, analyze the results using statistical methods (sampling, expected value, variance, standard deviation, normal distribution, z-scores), and produce a clear report and presentation with conclusions and recommendations.

Role

You are a member of the Learner Research Team responsible for planning, collecting, analyzing, and reporting on a real-life school issue (social media usage).

Audience

Your report and presentation will be delivered to:

Your classmates and teacher (primary audience)

School monitoring team or guidance counselor (secondary audience — optional)

Situation

Senior high students and school staff are concerned about social media use and its effects on study time and well-being. The school needs accurate, data-driven information to inform possible interventions. Your team must produce a reliable mini-study that follows good sampling practice and sound statistical analysis.

Product/Performance

Submit the following a proposal paper indicated the following information:

1. **Sampling Plan (document)** — research topic, population, sample size and justification, sampling method and short rationale, parameter vs statistic.
2. **Data Collection Sheet** — raw data (anonymized).
3. **Discrete Data Analysis (report + frequency table)** — frequency distribution, expected value (mean), variance, standard deviation, interpretation.
4. **Continuous Data Analysis (report + graph)** — normal curve sketch (or computer plot), z-score calculations for two chosen values, probabilities/percentiles from z-table, interpretation.
5. **Conclusion & Reflection (3–5 sentences)** — answer the three reflection prompts.
6. **Presentation (3–5 minutes)** — a concise slide or poster summarizing methods, key results (numbers & graph), conclusion & recommendation.
7. Include an appendix with the sampling instrument (questionnaire/consent note) and any calculation.

Standards & Success Criteria (Rubric / How performance will be judged)

Below are clear success criteria mapped to the rubric percentages. Use this as a checklist while you work.

Success Criteria (Checklist)

1. Topic, population, and sample size are clearly stated.
2. Sample size is justified (basic reasoning or formula).
3. Appropriate sampling method chosen and briefly explained.
4. Parameter and statistic are correctly identified for your study.
5. Discrete variable and continuous variable are correctly classified and justified.
6. Frequency table is complete and correct.
7. Mean (expected value), variance, and standard deviation are computed correctly.
8. Normal curve is sketched or plotted and labeled (mean, SD, chosen values).
9. Two z-scores correctly computed; corresponding probabilities/percentiles correctly read.

10. Interpretations are clear and link results to the research question.
11. Conclusion answers what the data showed, evaluates sampling method, and explains how statistics supported the conclusion.
12. Presentation is clear, neat, and within time limit.

Rubrics for Performance Task

Criteria	Excellent	Good	Fair	Needs Improvement
Sampling Plan 20%	Clear topic, population, sample size, method, explanation	Mostly clear; minor gaps	Incomplete or weak	Wrong/missing
Variable Classification 15%	Correct discrete & continuous with reasons	Correct but brief	Partly correct	Wrong/missing
Discrete Data Analysis 30%	Correct table, mean, variance, SD, clear interpretation	Minor errors	Several errors/unclear	Wrong/missing
Normal Distribution 25%	Correct z-scores, probabilities, curve, interpretation	Minor errors	Some correct, incomplete	Wrong/missing
Conclusion & Reflection 10%	Clear, thoughtful reflection	Adequate	Minimal/vague	Missing
Presentation 10%	Accurate, neat, organized	Mostly accurate; minor issues	Some errors; messy	Many errors